

Lecture 11: Quantum mechanics and molecular structure-2

Jin Xie (谢进), SPST, ShanghaiTech

Email: xiejin@shanghaitech.edu.cn

Classroom: Teaching Center 401

Wednesday/Friday: 10:15-11:00; 11:10-11:55

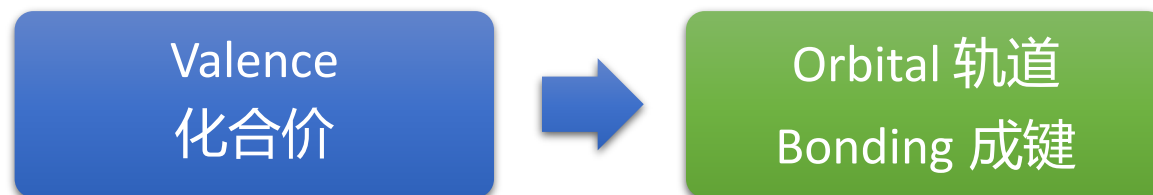
2025/11/14

Major points of last lecture

- **Quantum mechanics and molecular structure**
 1. Quantum Picture of the Chemical Bond
 2. Exact Molecular Orbitals (MO) for the Simplest Molecule H_2^+
 3. Molecular Orbital Theory and the Linear Combination of Atomic Orbitals Approximation for H_2^+
 4. Homonuclear Diatomic Molecules: First-Period Atoms

From Classical to Modern Chemistry

- The way of thinking



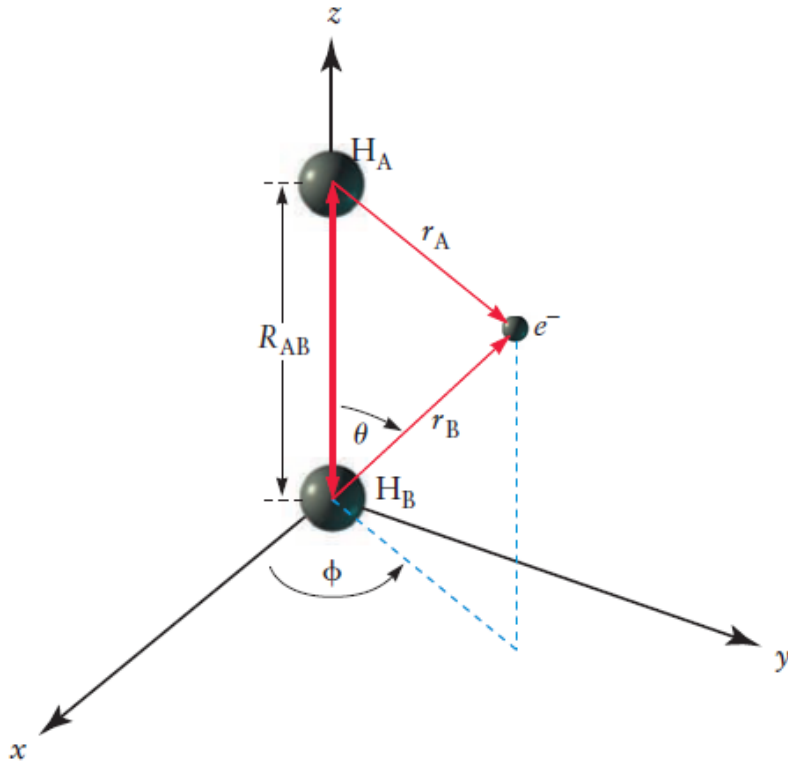
- The language



- Capability to analyze unfamiliar systems under theoretical guidance

Exact Molecular Orbitals (MO) for the Simplest Molecule H_2^+

Solution of Schrödinger equation: $\psi_{\text{mol}}(\mathbf{R}_{AB}, r_A, r_B, \phi)$, a function of all four of these coordinates.



- symmetry of the system has changed from **spherical** to **ellipsoidal**.
- converted the system into the so-called three-body problem, for which there is **no exact solution**.
- **Born-Oppenheimer approximation** that treats the nuclear and electronic motions almost as if they were independent.
- This method allows us to solve the Schrödinger equation exactly and provides an solution $\psi_{\text{el}}(r_A, r_B, \phi; R_{AB})$

Born-Oppenheimer Approximation

The basic idea of the **Born-Oppenheimer approximation (玻恩-奥本海默近似)**: because the nuclei are so much more massive than the electrons. **Slow Nuclei, Fast Electrons.**

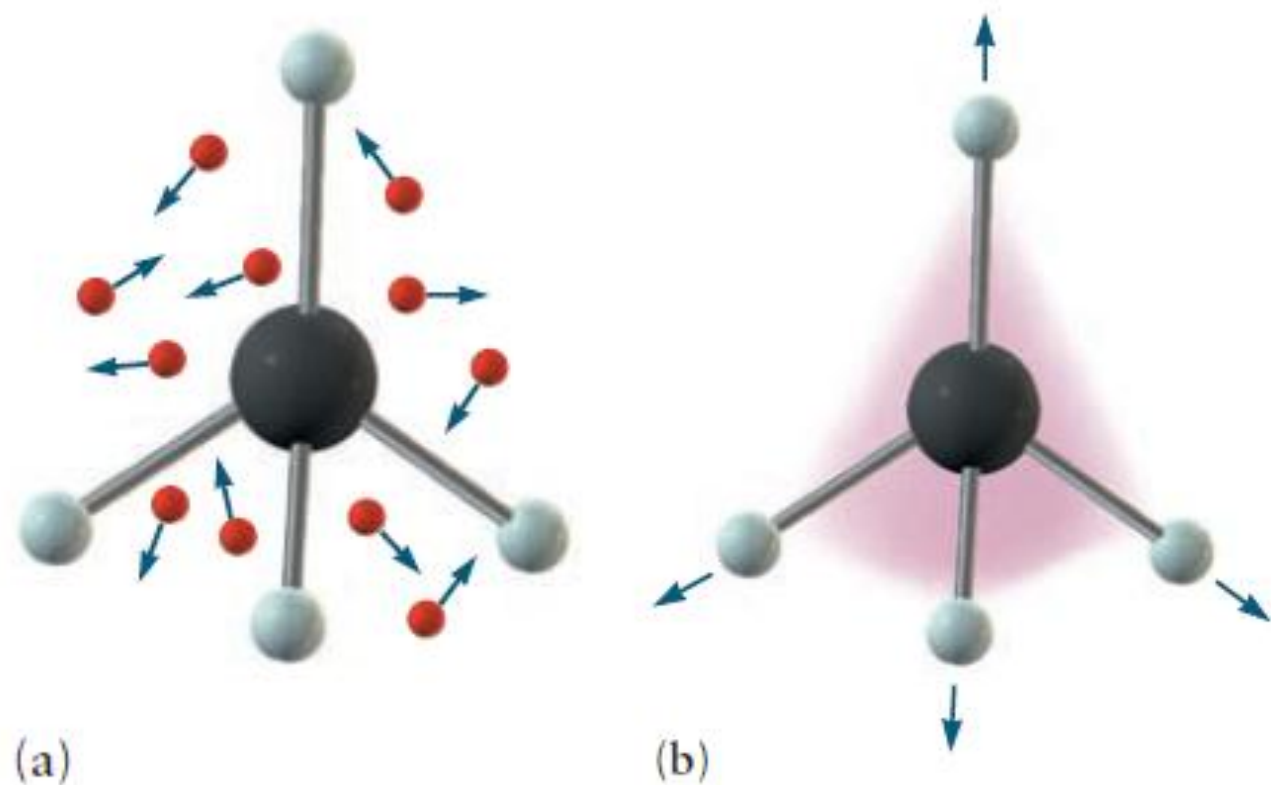
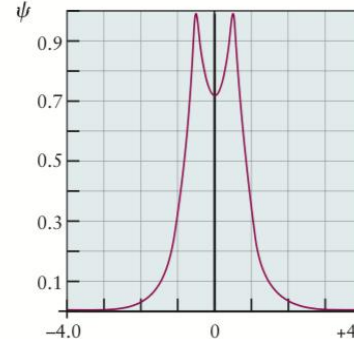
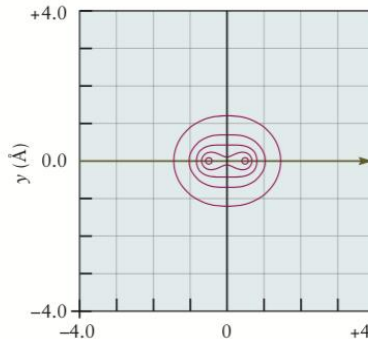
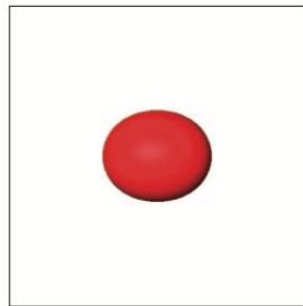
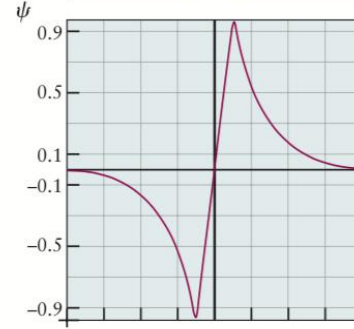
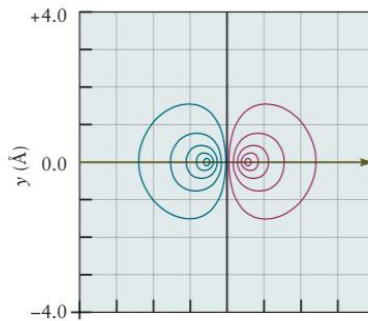
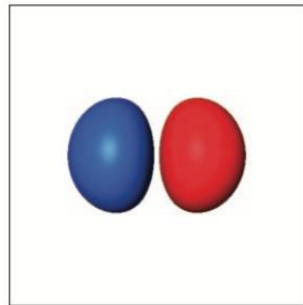
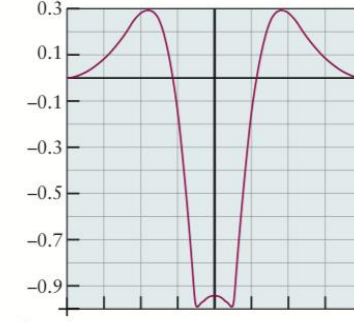
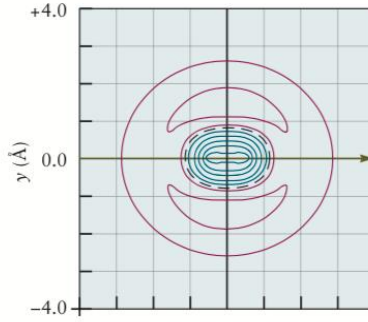
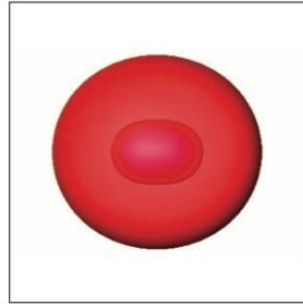
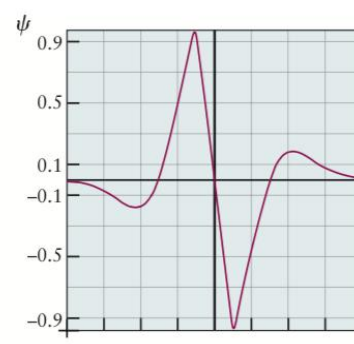
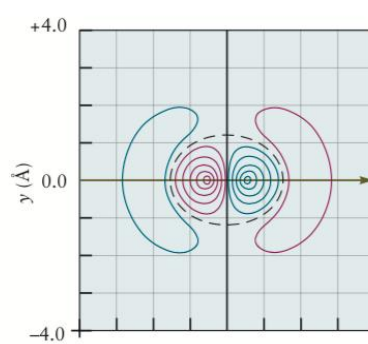
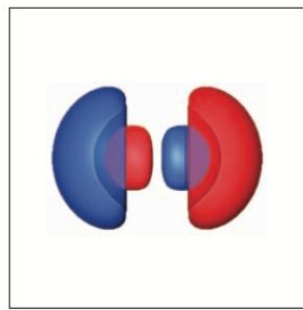


FIGURE 6.3 Schematic representation of the Born–Oppenheimer approximation for methane. (a) The nuclei are assumed to be fixed at positions corresponding to the equilibrium bond distances, while the electrons move rapidly around them. The wave functions and allowed energy values are calculated for the electrons at many different positions of the nuclei. (b) The allowed energy values calculated in (a), with the internuclear repulsions added in, are viewed as the effective potential energy functions that govern the relative motions of the nuclei in the molecule, as occurs in vibrations. The multidimensional effective potential energy function is suggested here as a cloud.

$$1\sigma_g < 1\sigma_u^* < 2\sigma_g < 2\sigma_u^*$$

- The **energy index integer**: tracks the relative energies of the wave functions of each symmetry type. (analogous to the principal quantum number n for atomic orbitals)
- As the index increases within each symmetry type, the number of nodes along the internuclear axis increases.
- The subscript **g or u** describes how the wave function changes as we invert our point of observation through the center of the molecule.
- The sign of the wave function is the **same** at the point (x, y, z) and at the point $(-x, -y, -z)$, it is called **symmetric** and labeled **g**
- If the sign of the wave function is **opposite** at these two points, it is called **antisymmetric** and labeled **u**



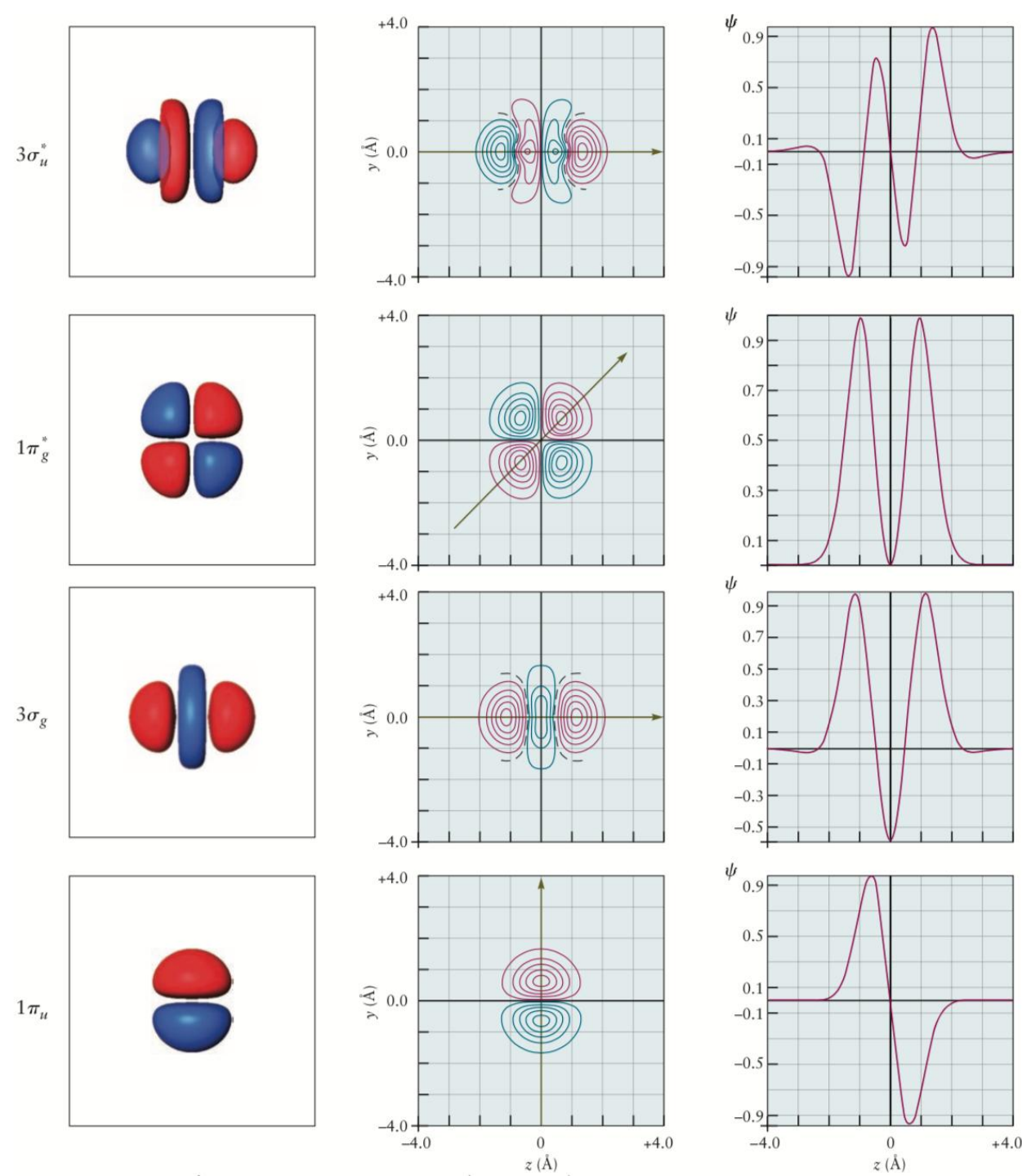
(a) Isosurfaces

(b) Contour plots

(c) Line scans

$$1\pi_u < 3\sigma_g < 1\pi_g^* < 3\sigma_u^*$$

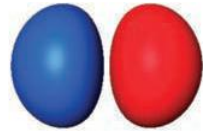
- The Greek letter: how the amplitude of the wave function is distributed around the internuclear axis.
- The letter σ : the amplitude has **cylindrical symmetry** around the axis.
- The letter π : the wave function has **a nodal plane that contains the internuclear axis**.



- asterisk “ * ” denotes an **antibonding orbital**.
- If the wave function changes sign upon reflection, it receives the **superscript ***. Each wave function labeled * has **a node on the internuclear axis at the center of the molecule**.



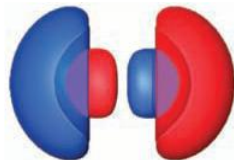
$1\sigma_g$



$1\sigma_u^*$



$2\sigma_g$

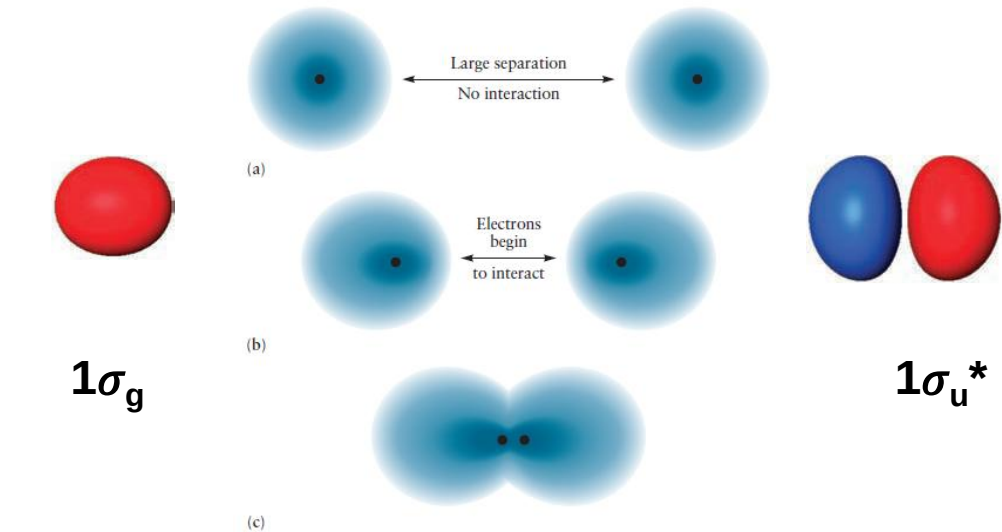


$2\sigma_u^*$

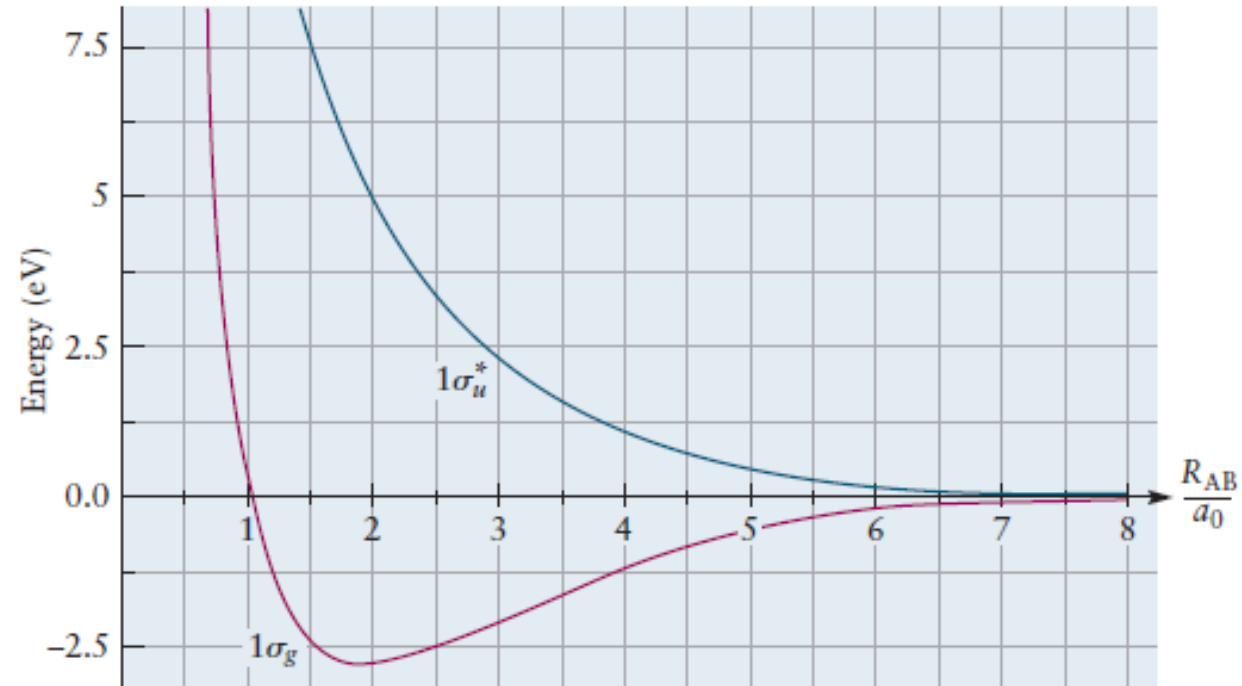
1 nodal plane **in the middle of** the bond.

Nature of the chemical bond in H_2^+

- $1\sigma_g$ is called a bonding molecular orbital.
- $1\sigma_u^*$ is called a antibonding molecular orbital.
- The $1\sigma_u^*$ orbital shows zero electron density halfway between the nuclei and increased energy.



$$\psi_{\text{mol}}(R_{\text{AB}}, r_{\text{A}}, r_{\text{B}}, \phi) \approx \psi_{\text{el}}(r_{\text{A}}, r_{\text{B}}, \phi; R_{\text{AB}}) \times \psi_{\text{nuc}}(R_{\text{AB}})$$



Linear Combination of Atomic Orbitals (LCAO) method

Linear Combination of Atomic Orbitals (LCAO, 原子轨道线性组合) : selecting sums and differences (linear combinations) of atomic orbitals to generate the best **approximation** to each type of molecular orbital.

Approximation for H_2^+

- Electron is close to nucleus A: not very different from that in an isolated hydrogen atom. The ground-state wave function for the electron near A should therefore resemble a 1s atomic wave function φ_{1s}^A .
- Near B: the wave function should resemble φ_{1s}^B .
- An **approximate MO** for H_2^+ : a linear combination of two atomic orbitals, obtained by adding or subtracting the two: with coefficients whose values depend on R_{AB} :

$$\psi_{\text{MO}} = C_A(R_{AB})\varphi_{1s}^A \pm C_B(R_{AB})\varphi_{1s}^B$$

Linear Combination of Atomic Orbitals (LCAO) method

New symbols σ_{g1s} and σ^*_{u1s} for the approximate MOs:

- (1) distinguish them from the exact MOs;
- (2) indicate explicitly the AOs from which they were constructed.

$$1\sigma_g \approx \sigma_{g1s} = C_g[\varphi^A_{1s} + \varphi^B_{1s}]$$

$$1\sigma^*_u \approx \sigma^*_{u1s} = C_u[\varphi^A_{1s} - \varphi^B_{1s}]$$

T A B L E 6.1	
Molecular Orbitals for Homonuclear Diatomic Molecules	
Exact MO Notation	LCAO MO Notation
$1\sigma_g$	σ_{g1s}
$1\sigma^*_u$	σ^*_{u1s}
$2\sigma_g$	σ_{g2s}
$2\sigma^*_u$	σ^*_{u2s}
$1\pi_u$	$\pi_{u2p_x} \pi_{u2p_y}$
$3\sigma_g$	σ_{g2p_z}
$1\pi^*_g$	$\pi^*_{g2p_x} \pi^*_{g2p_y}$
$3\sigma^*_u$	$\sigma^*_{u2p_z}$

Linear Combination of Atomic Orbitals (LCAO) method

The distribution of **electron probability density**: squaring each of the approximate MOs:

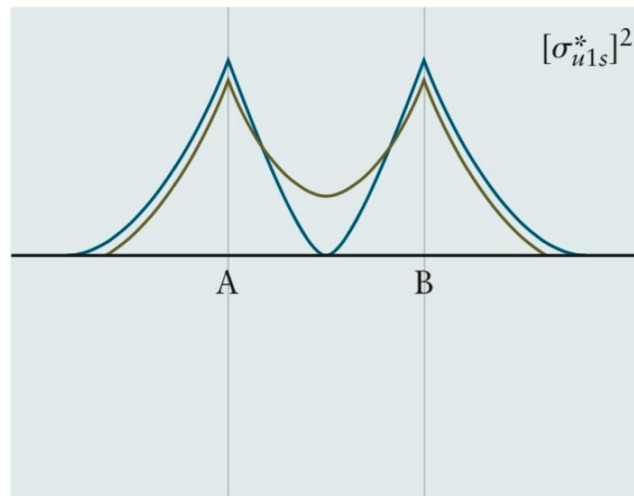
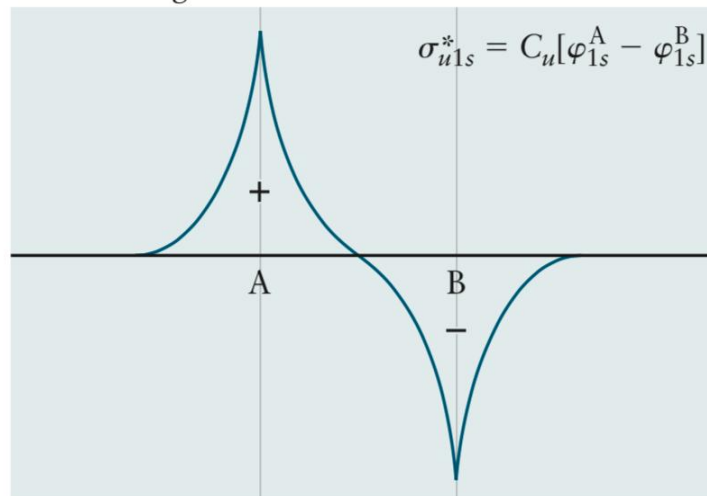
$$[\sigma_{g1s}]^2 = C_g^2 [(\varphi_{1s}^A)^2 + (\varphi_{1s}^B)^2 + 2\varphi_{1s}^A \varphi_{1s}^B]$$

$$[\sigma_{u1s}^*]^2 = C_u^2 [(\varphi_{1s}^A)^2 + (\varphi_{1s}^B)^2 - 2\varphi_{1s}^A \varphi_{1s}^B]$$

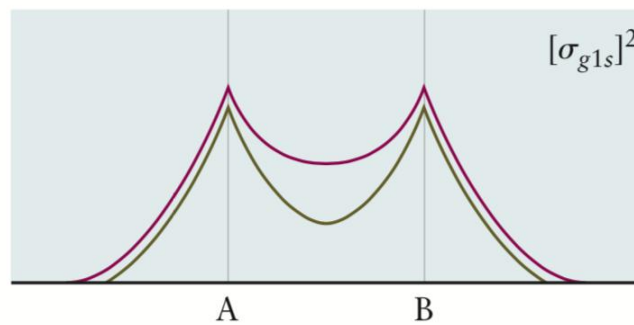
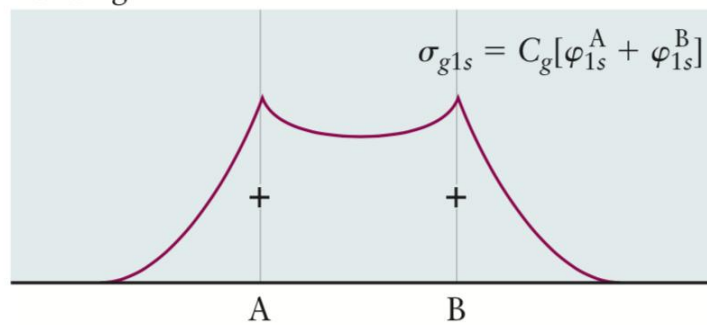
The probability distribution for a non-interacting (n.i.) system (averaging the probabilities for $H_A^+ + H_B$ and $H_A + H_B^+$):

$$\psi_{n.i.}^2 = C_3^2 [(\varphi_{1s}^A)^2 + (\varphi_{1s}^B)^2]$$

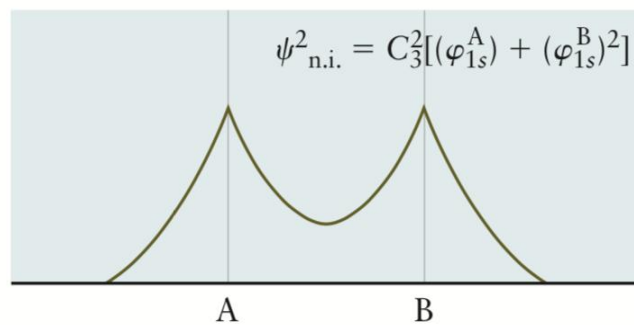
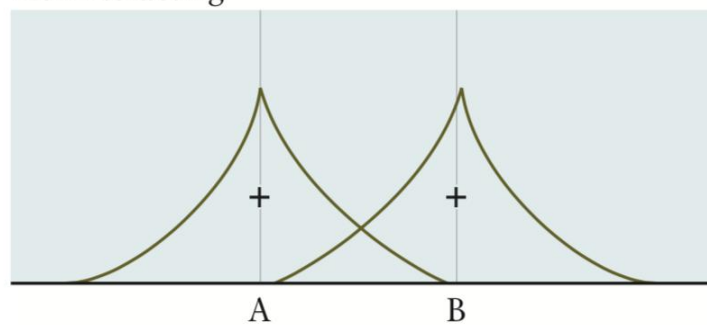
Antibonding



Bonding

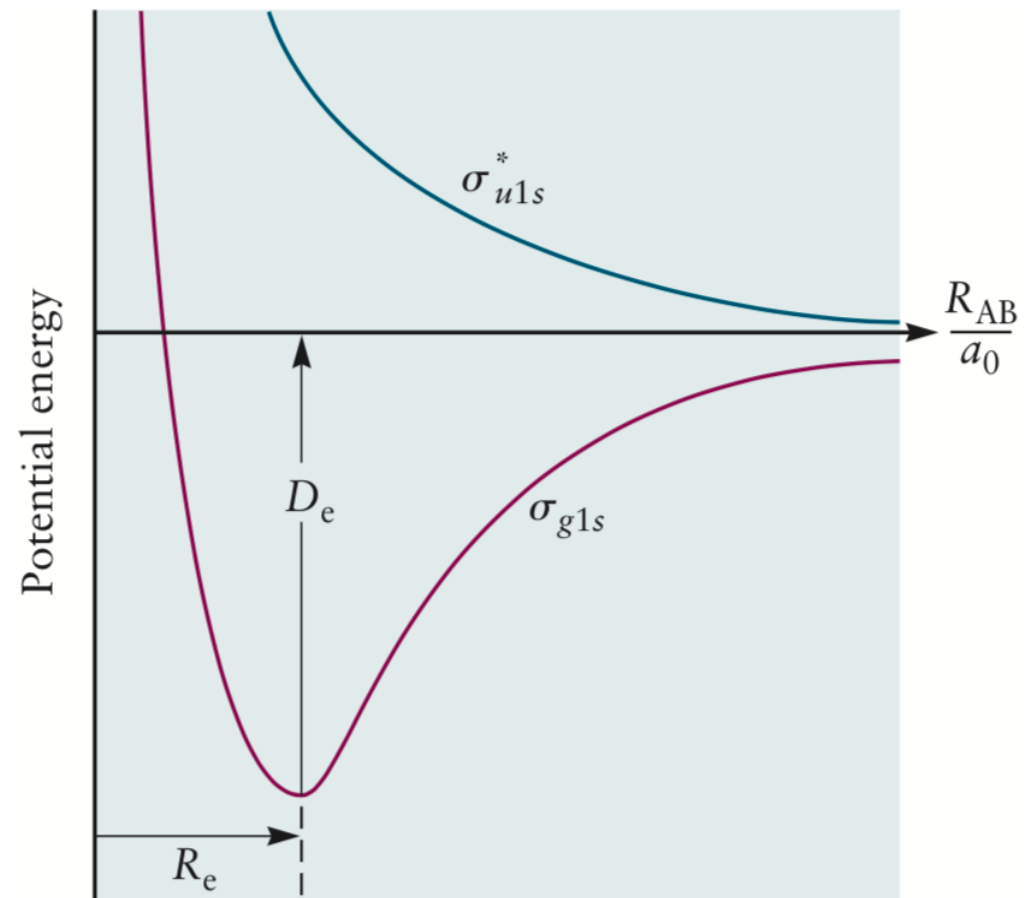


Noninteracting

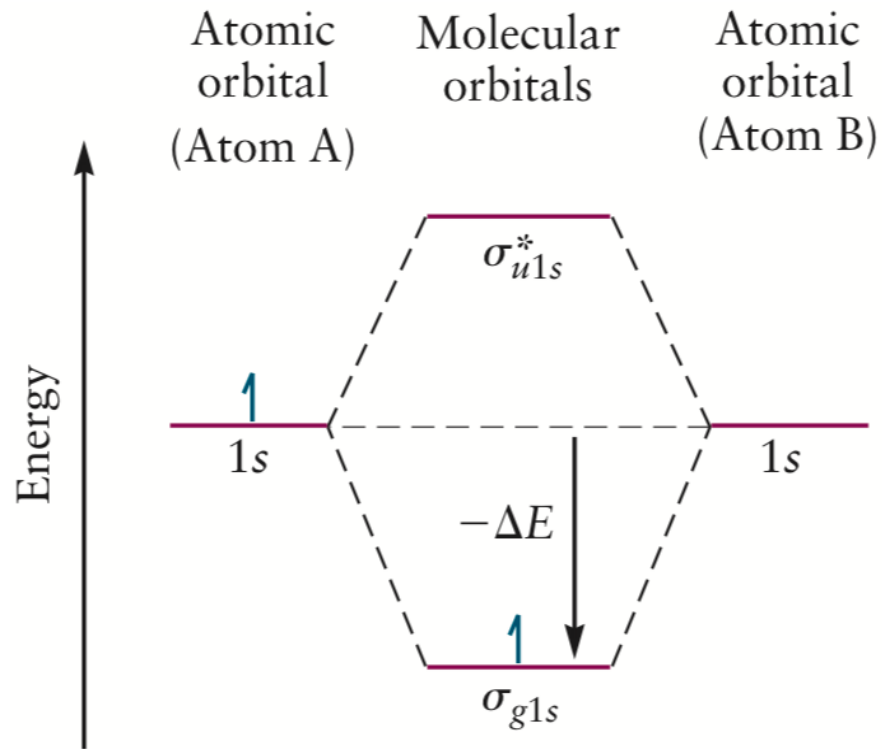


Wave functions

Electron densities



Linear Combination of Atomic Orbitals (LCAO) method



Correlation diagram (轨道相关图): the energy-level diagram within the LCAO approximation

Two 1s AOs have been combined to give a σ_{g1s} MO with energy lower than the AOs and a σ_{u1s}^* MO with energy higher than the AOs.

FIGURE 6.10 Correlation diagram for H_2^+ in the linear combination of atomic orbitals (LCAO) approximation. The bonding orbital is stabilized relative to the noninteracting system by the energy difference $-\Delta E$.

Homonuclear Diatomic Molecules: First-Period Atoms

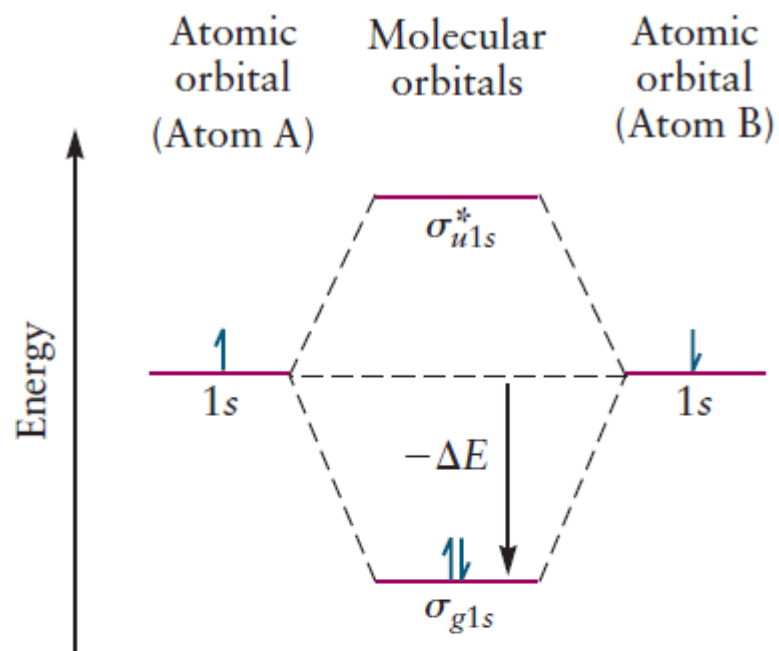


FIGURE 6.11 Correlation diagram for first-period diatomic molecules. Blue arrows indicate the electron filling for the H_2 molecule. All of the atomic electrons are pooled and used to fill the molecular orbitals following the aufbau principle. In the molecules, electrons are no longer connected to any particular atom.

Combine the **LCAO method** with an **aufbau principle**:
Electrons available from the two atoms are “fed” into the molecular orbitals, starting with the MO of lowest energy.

Homonuclear Diatomic Molecules: First-Period Atoms

First-Period Atoms: the LCAO approximation can be applied in this same way to He_2^+ and He_2

- **One change:** the electrons in He_2^+ and He_2 experience a potential close to that in a He atom (**Use He 1s AOs instead of H 1s AOs**).

The equations for the MOs:

$$\sigma_{g1s} = C_g[\varphi_{1s}^A(\text{He}) + \varphi_{1s}^B(\text{He})]$$

$$\sigma_{u1s}^* = C_u[\varphi_{1s}^A(\text{He}) - \varphi_{1s}^B(\text{He})]$$

$$\sigma_{g1s} = C_g[1s^A + 1s^B]$$

A simpler notation:

$$\sigma_{u1s}^* = C_u[1s^A - 1s^B]$$

Homonuclear Diatomic Molecules: First-Period Atoms

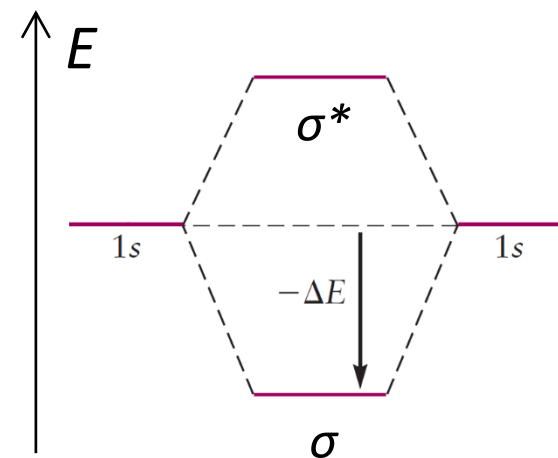
- In bonding MOs: bond formation arises from the sharing of electrons. The average electron density is greatest between the nuclei and tends to pull them together.
- Electrons that are shared in an antibonding MO tend to force the nuclei apart, reducing the bond strength.

$$\text{bond order} = \frac{1}{2} (\text{number of electrons in bonding molecular orbitals} - \text{number of electrons in antibonding molecular orbitals})$$

Homonuclear Diatomic Molecules: First-Period Atoms

systematic procedure: **three steps**

1. Form linear combinations of the **minimum basis set of AOs to generate MOs**. The total number of MOs formed in this way must equal the number of AOs used.
2. Arrange the MOs in order **from lowest to highest energy**.
3. **Put in electrons** (at most two electrons per MO), **starting from the orbital of lowest energy**. Apply Hund's rules when appropriate.



Molecular orbitals *do not* change:

- Total number of **orbitals**
- Total **energy** of all orbitals
- Total number of **electrons**

Major points of today's lecture

- **Quantum mechanics and molecular structure**
 1. Quantum Picture of the Chemical Bond
 2. Exact Molecular Orbitals (MO) for the Simplest Molecule H_2^+
 3. Molecular Orbital Theory and the Linear Combination of Atomic Orbitals Approximation for H_2^+
 4. Homonuclear Diatomic Molecules: First-Period Atoms
 5. Homonuclear Diatomic Molecules: Second-Period Atoms
 6. Heteronuclear Diatomic Molecules
 7. Summary Comments for the LCAO Method and Diatomic Molecules

Homonuclear Diatomic Molecules: Second-Period Atoms

These MOs as a linear combination of the 1s, 2s, and 2p orbitals with proper coefficients.

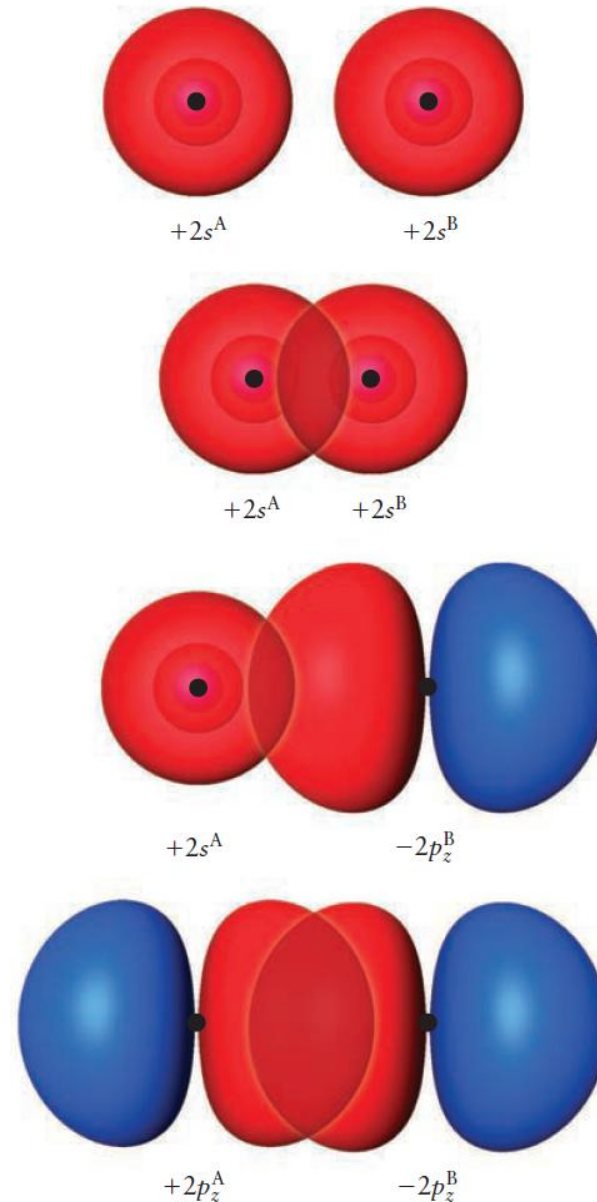
1. Two atomic orbitals contribute significantly to bond formation only if their **atomic energy levels are very close to one another**.
2. Two atomic orbitals on different atoms contribute significantly to bond formation only if they **overlap significantly**.

Homonuclear Diatomic Molecules: Second-Period Atoms

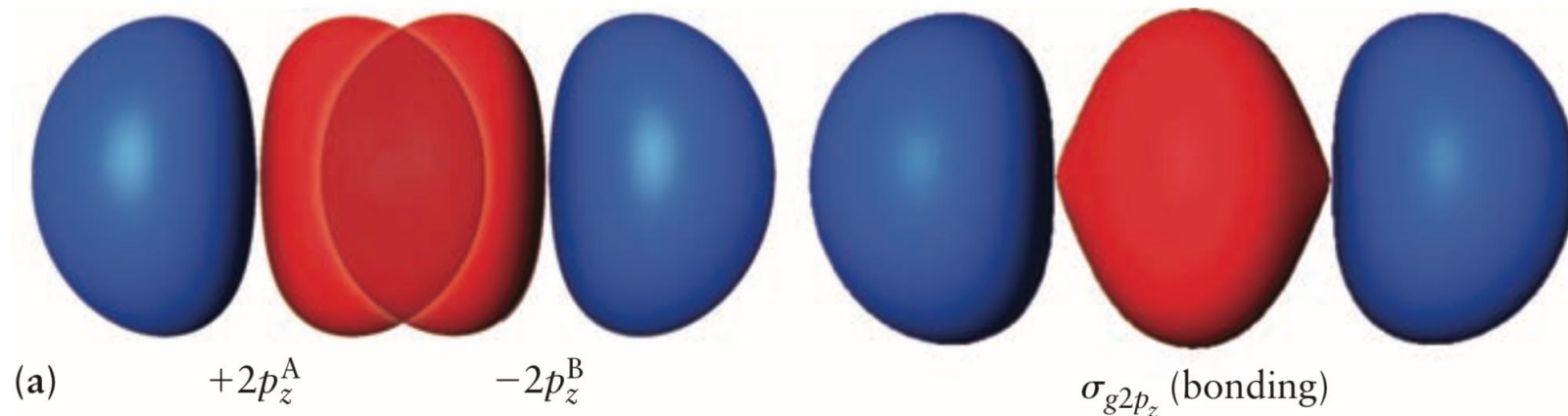
2s AOs of the two atoms, giving a σ_{g2s} bonding orbital and a σ_{u2s}^* antibonding orbital

$$\sigma_{g2s} = C_g[2s^A + 2s^B]$$

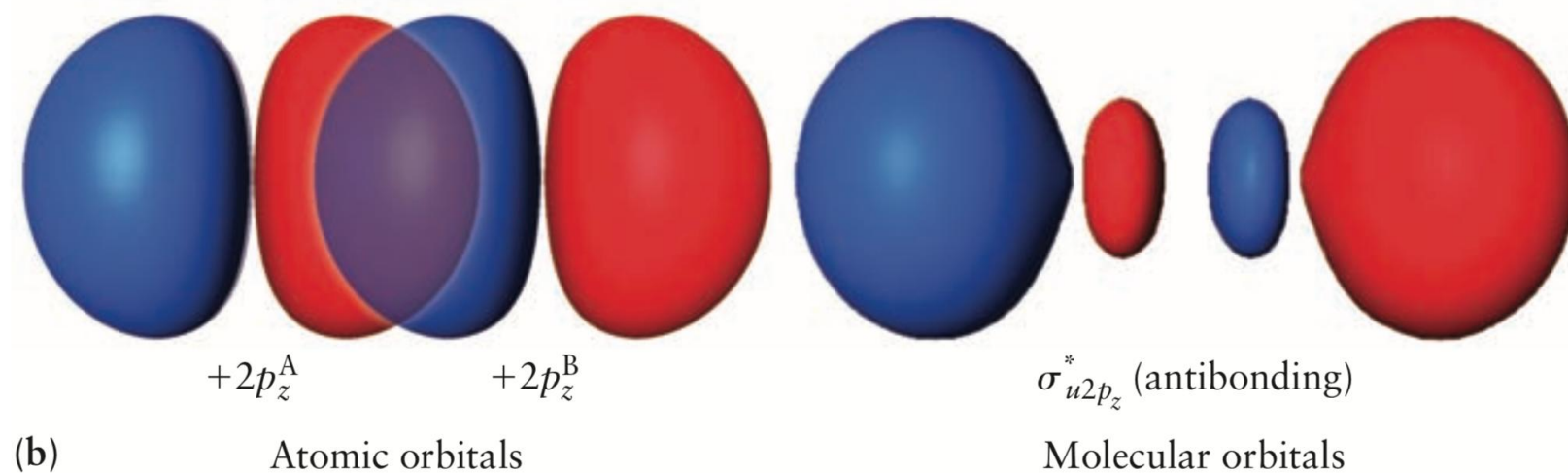
$$\sigma_{u2s}^* = C_u[2s^A - 2s^B]$$



Homonuclear Diatomic Molecules: Second-Period Atoms



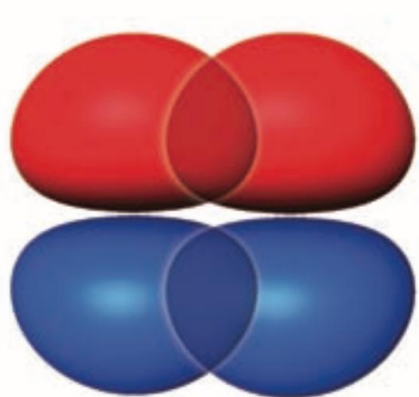
2p orbitals form different MOs depending on whether they are **parallel or perpendicular to the internuclear (bond) axis**



$$\sigma_{g2p_z} = C_g[2p_z^A - 2p_z^B]$$

$$\sigma_{u2p_z}^* = C_u[2p_z^A + 2p_z^B]$$

Homonuclear Diatomic Molecules: Second-Period Atoms

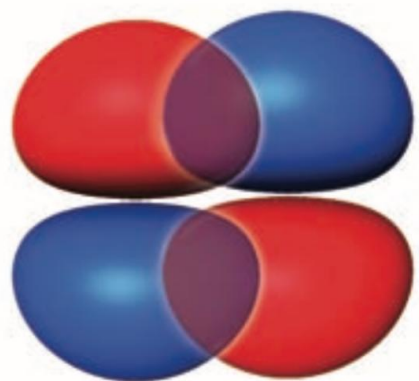


$+2p_x^A \quad +2p_x^B$
(a)



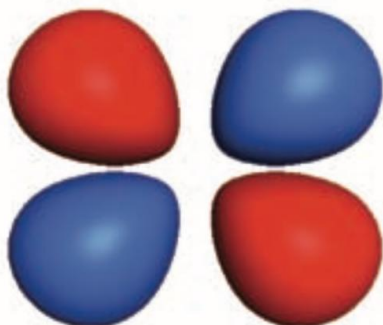
π_{u2p_x} (bonding)

These orbitals have **a nodal plane that contains the internuclear axis** (in this case, the y-z plane) and are designated by p rather than s. In the same way, π_{u2py} and π_{g2py}^* orbitals can be formed



$+2p_x^A \quad -2p_x^B$

Atomic orbitals
(b)



$\pi_{g2p_x}^*$ (antibonding)

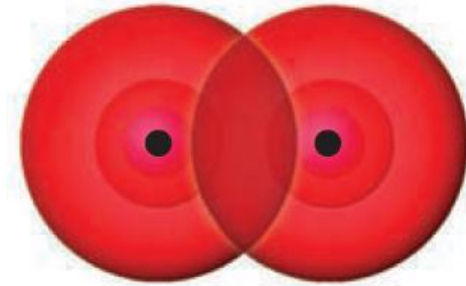
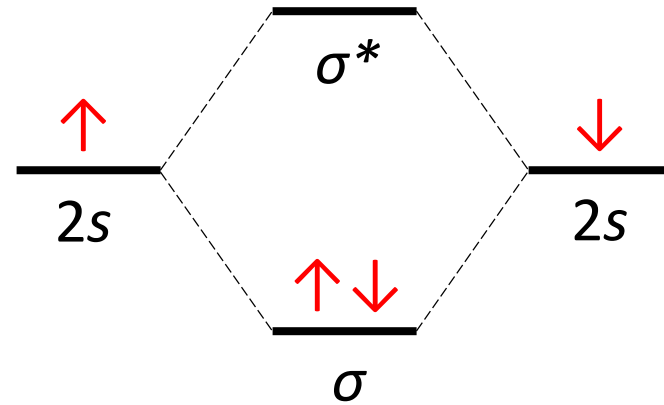
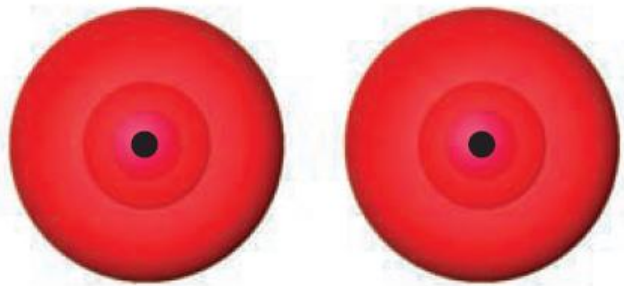
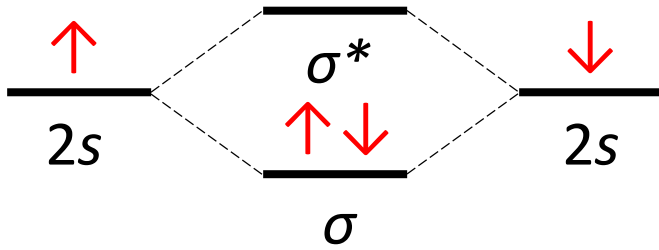
Molecular orbitals

$$\pi_{u2p_x} = C_u[2p_x^A + 2p_x^B]$$

$$\pi_{g2p_x}^* = C_g[2p_x^A - 2p_x^B]$$

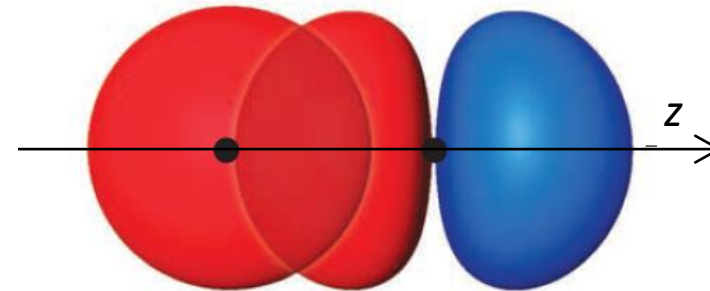
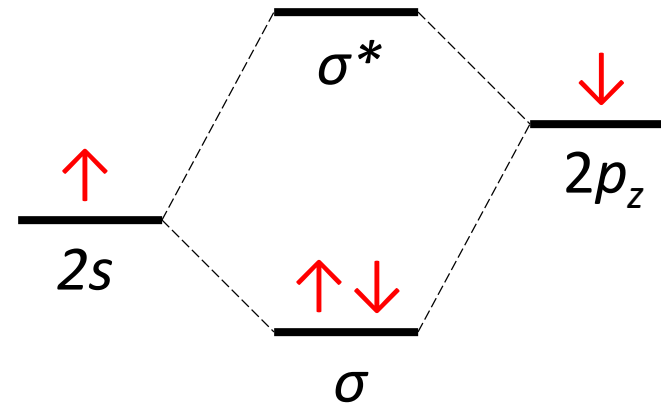
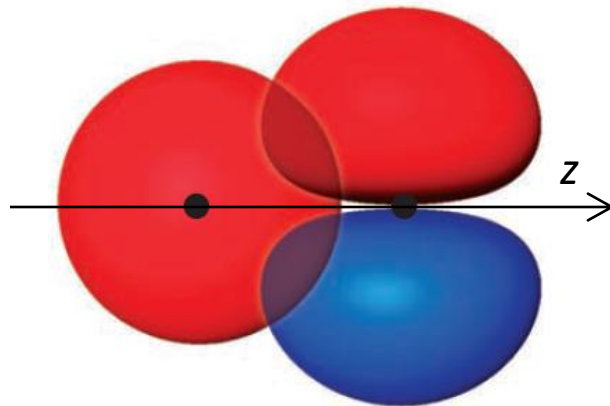
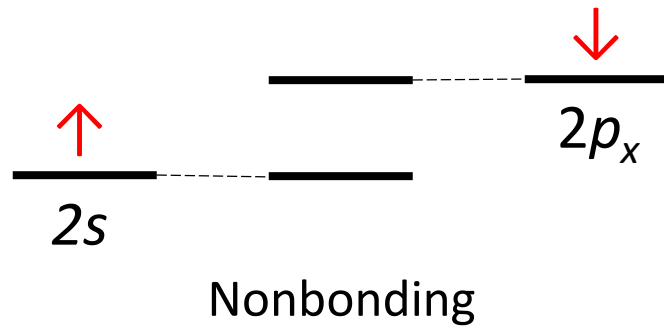
Factors for Effective Bonding

① Maximum orbital overlap



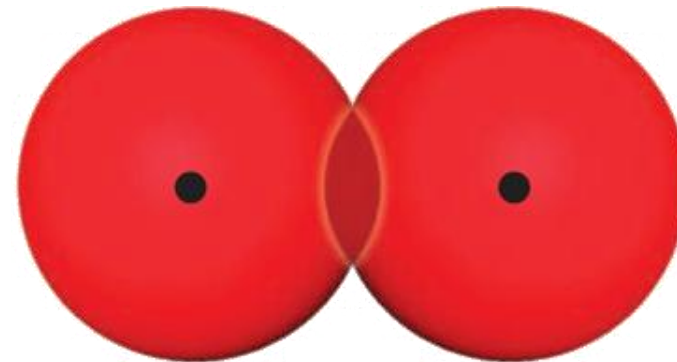
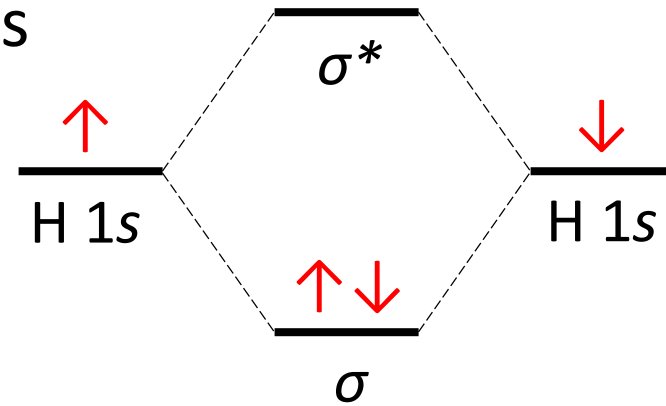
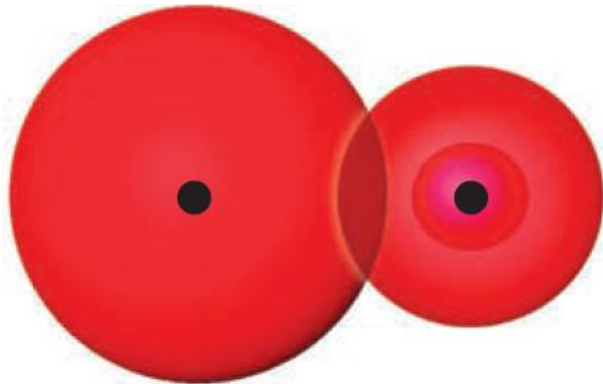
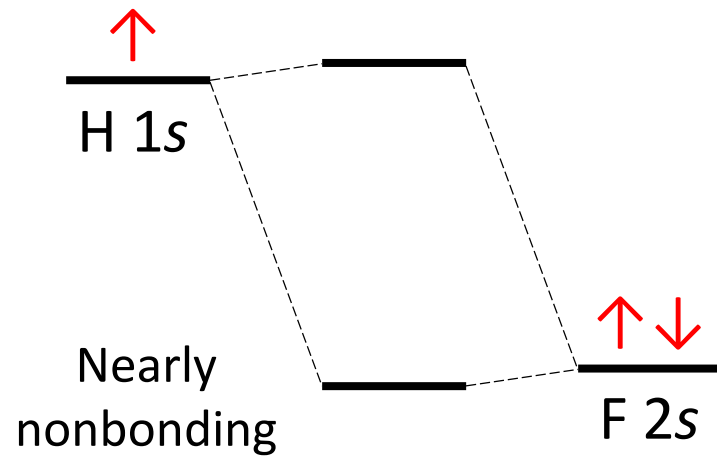
Factors for Effective Bonding

② Compatible orbital symmetries

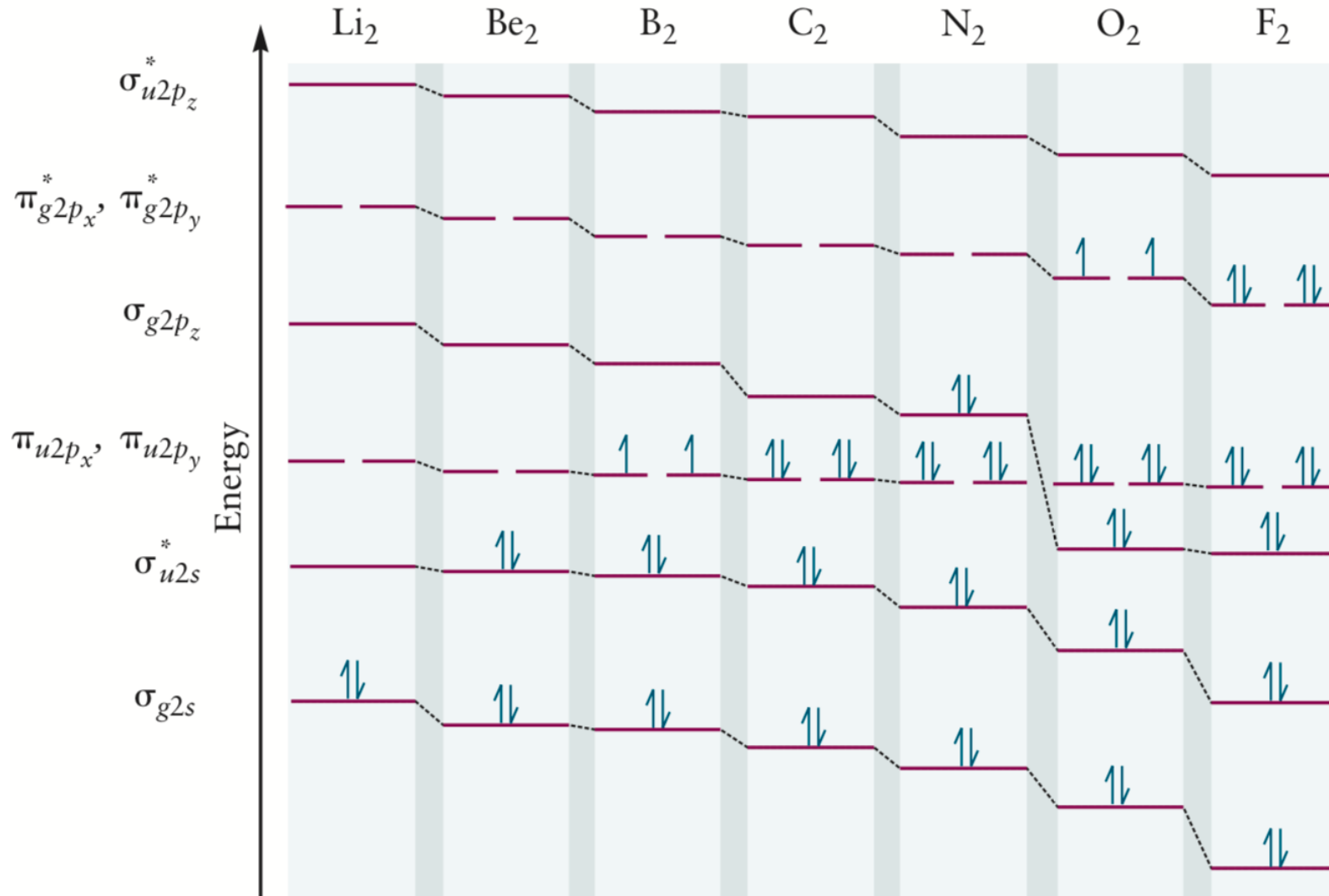


Factors for Effective Bonding

③ Approximate atomic orbital energies



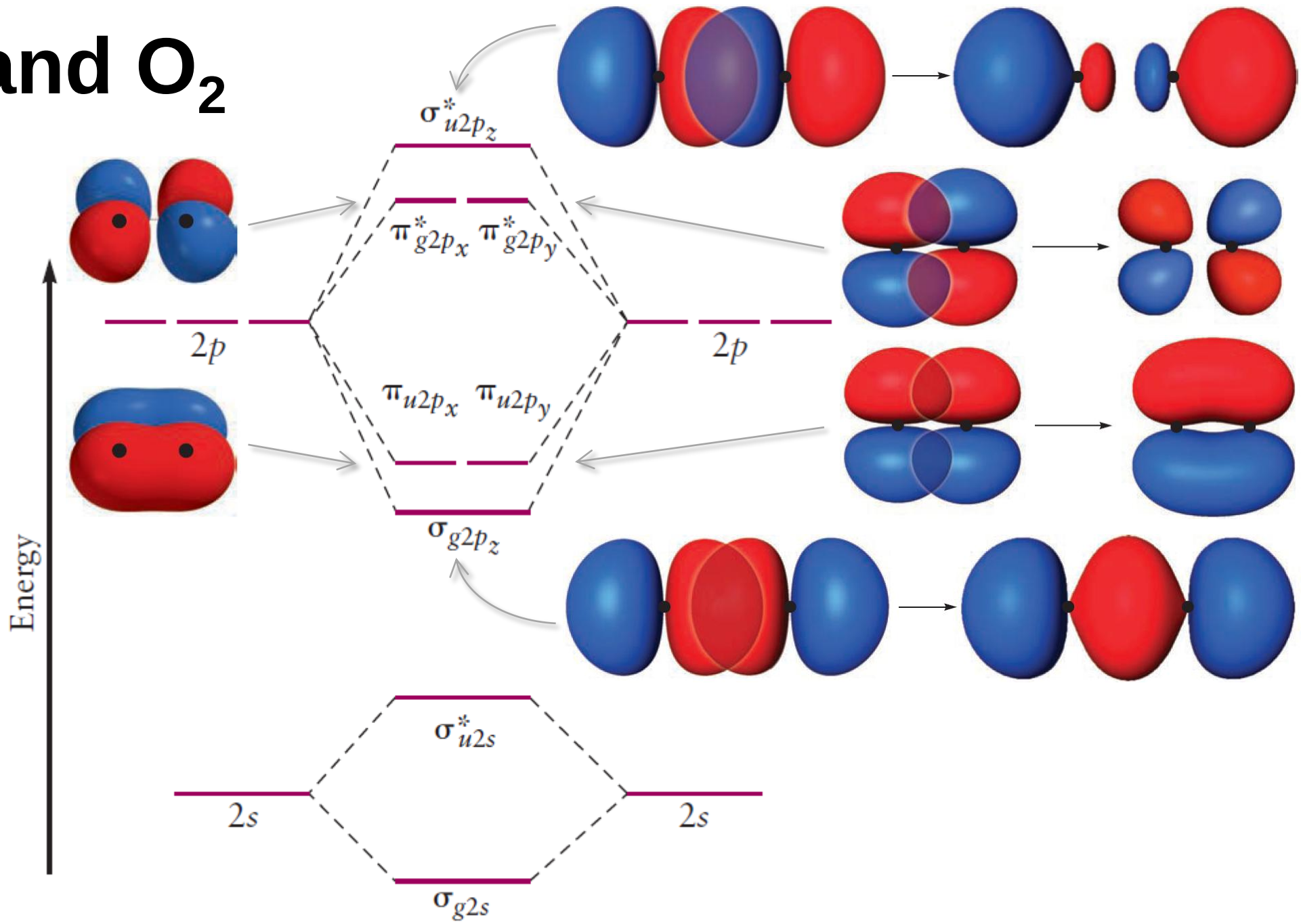
Homonuclear Diatomic Molecules: Second-Period Atoms



increased nuclear charge (Z_{eff}) felt by the electrons in the heavier elements.

The electrons have been placed in MOs according to the Aufbau principle.

F₂ and O₂



Sept., 1924

MAGNETISM OF OXYGEN

2027

[CONTRIBUTION FROM THE CHEMICAL LABORATORY OF THE UNIVERSITY OF CALIFORNIA]

THE MAGNETISM OF OXYGEN AND THE MOLECULE O_4

BY GILBERT N. LEWIS

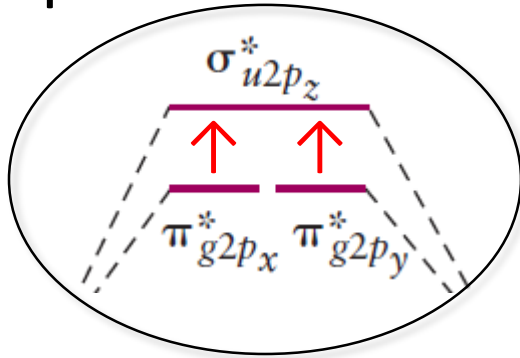
RECEIVED JUNE 25, 1924

PUBLISHED SEPTEMBER 5, 1924

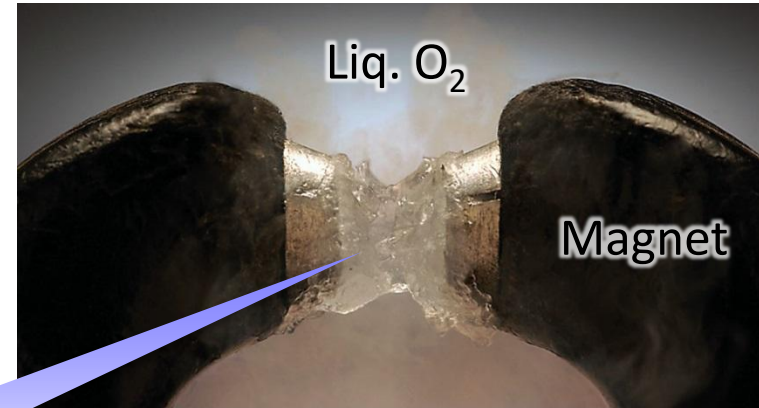
One of the most characteristic properties of oxygen is its pronounced paramagnetism. This distinguishes molecular oxygen so sharply from other compounds in which oxygen is assumed to be tied by a double bond that I have recently concluded¹ that there is no double bond in the O_2 molecule. Rather I have assumed that the two electrons which have been supposed to constitute the second bond are actually separate so that each atom has an odd electron, according to the formula $\overset{\cdot\cdot}{\underset{\cdot\cdot}{\text{O}}}:\overset{\cdot\cdot}{\underset{\cdot\cdot}{\text{O}}}:$. Indeed, oxygen has the chemical properties of a molecule with odd electrons, for it has an appreciable color, and a much higher reactivity than has commonly been ascribed to it.

O₂: Unusual Properties

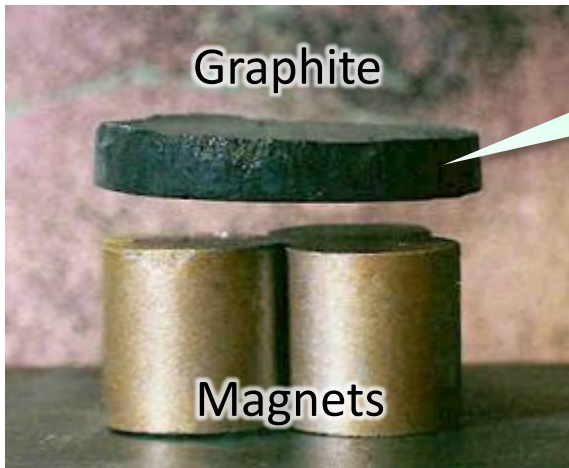
O₂ is paramagnetic due to its two unpaired electron spins.



Spinning electrons

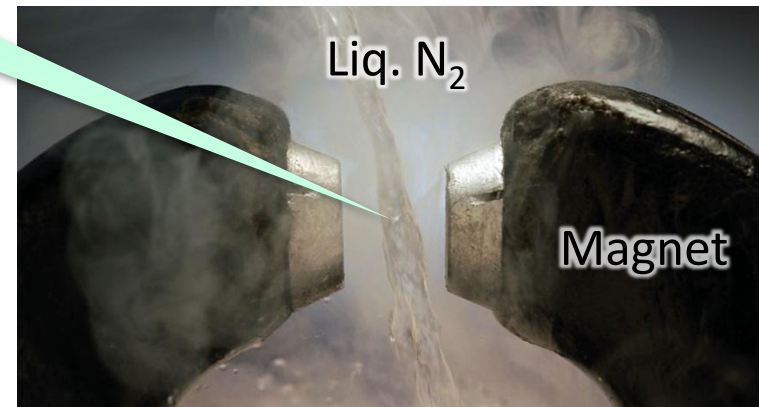


Paramagnetic 顺磁性

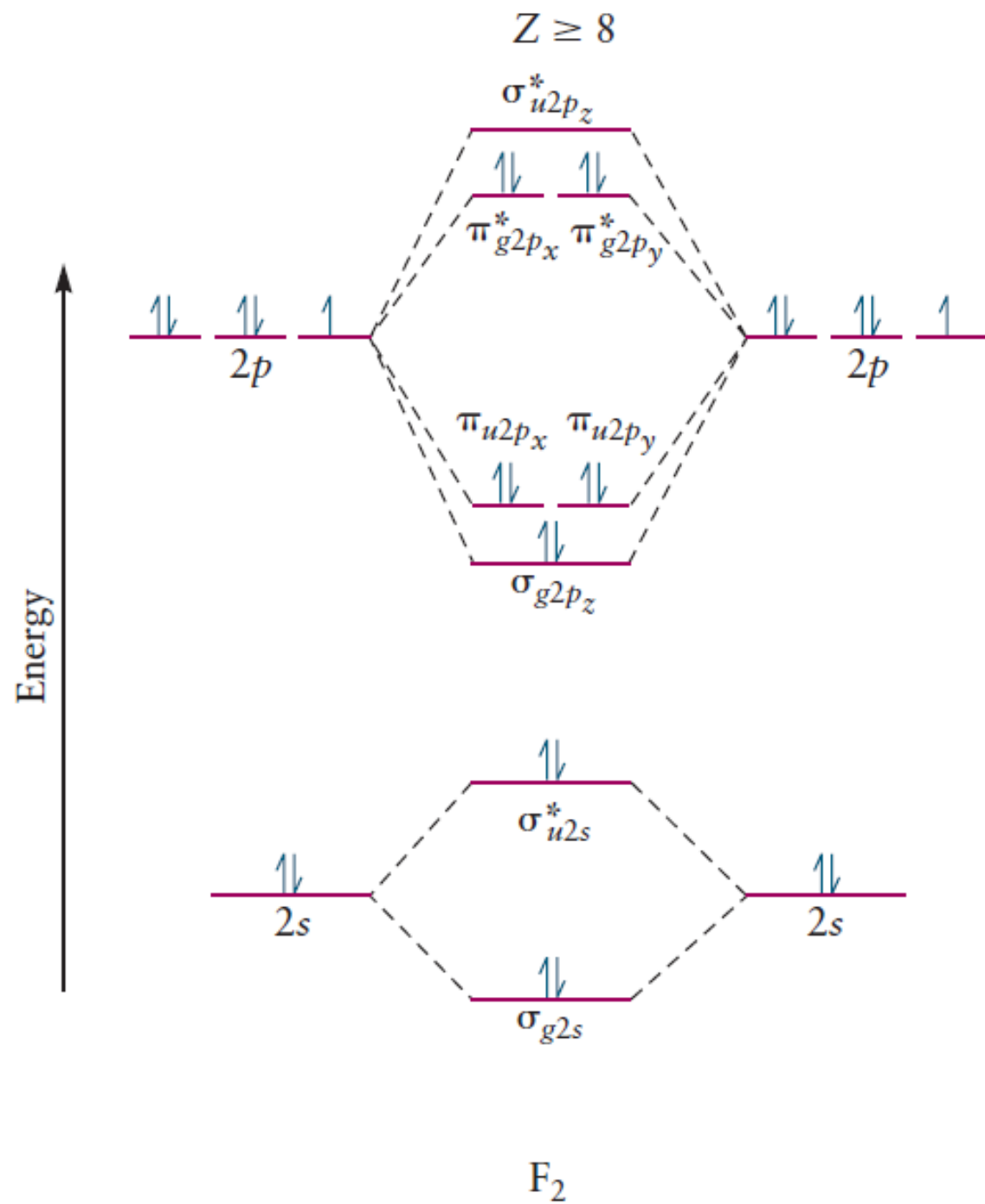


Orbiting electrons

Diamagnetic 反 (抗、逆) 磁性



Weakly diamagnetic



Second-Period Elements

Orbital configuration and energy (in eV)

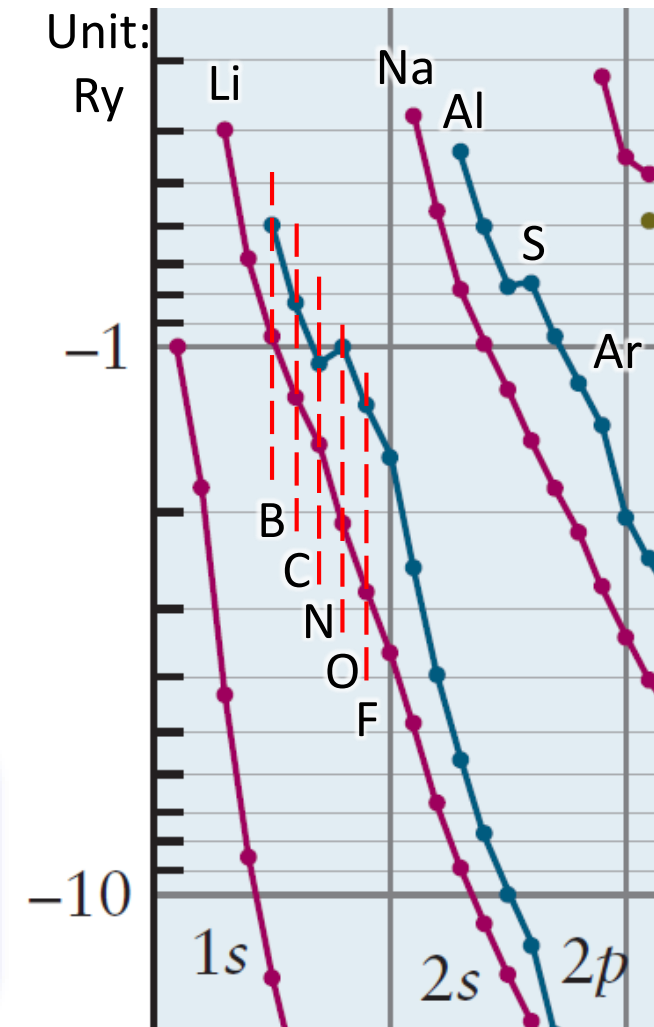
	B	C	N	O	F
2s	↑↓	↑↓	↑↓	↑↓	↑↓
	-14.1	-17.8	-25.4	-32.3	-40.2
2p	↑	↑ ↑	↑ ↑ ↑	↑ ↓ ↑ ↑	↑ ↓ ↑ ↓ ↑
	-8.3	-11.3	-14.5	-13.6	-17.4
2p - 2s	5.8	6.5	10.9	18.8	22.8

Small difference
 Large difference

s-p mixing 混合

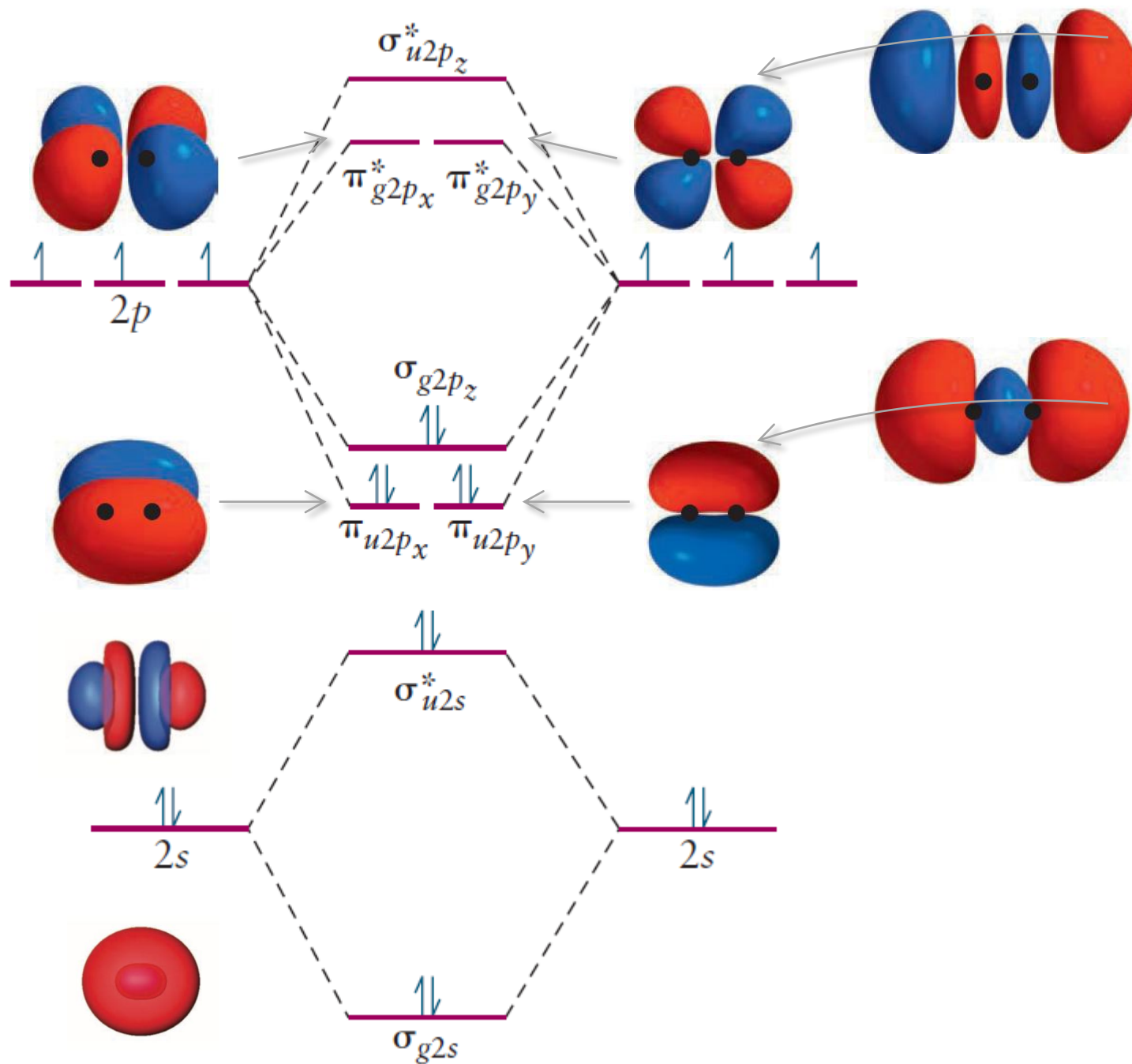
$\sigma_g(2s)$ interacts with $\sigma_g(2p_z)$ (both σ and g)
 $\sigma_u^*(2s)$ interacts with $\sigma_u^*(2p_z)$ (both σ and u)

Fig. 5.26, OGB8 p.210

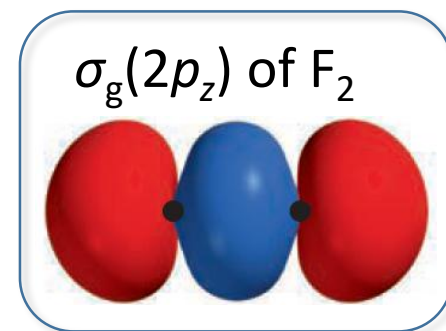


N₂

Energy ↑



VS



Homonuclear Diatomic Molecules: Second-Period Atoms

T A B L E 6.3 Molecular Orbitals of Homonuclear Diatomic Molecules

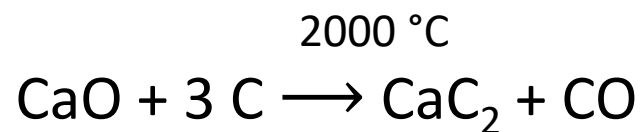
Species	Number of Valence Electrons	Valence Electron Configuration	Bond Order	Bond Length (Å)	Bond Energy (kJ mol ⁻¹)
H ₂	2	$(\sigma_{g1s})^2$	1	0.74	431
He ₂	4	$(\sigma_{g1s})^2(\sigma_{u1s}^*)^2$	0		
Li ₂	2	$(\sigma_{g2s})^2$	1	2.67	105
Be ₂	4	$(\sigma_{g2s})^2(\sigma_{u2s}^*)^2$	0	2.45	9
B ₂	6	$(\sigma_{g2s})^2(\sigma_{u2s}^*)^2(\pi_{u2p})^2$	1	1.59	289
C ₂	8	$(\sigma_{g2s})^2(\sigma_{u2s}^*)^2(\pi_{u2p})^4$	2	1.24	599
N ₂	10	$(\sigma_{g2s})^2(\sigma_{u2s}^*)^2(\pi_{u2p})^4(\sigma_{g2p_z})^2$	3	1.10	942
O ₂	12	$(\sigma_{g2s})^2(\sigma_{u2s}^*)^2(\sigma_{g2p_z})^2(\pi_{u2p})^4(\pi_{g2p}^*)^2$	2	1.21	494
F ₂	14	$(\sigma_{g2s})^2(\sigma_{u2s}^*)^2(\sigma_{g2p_z})^2(\pi_{u2p})^4(\pi_{g2p}^*)^4$	1	1.41	154
Ne ₂	16	$(\sigma_{g2s})^2(\sigma_{u2s}^*)^2(\sigma_{g2p_z})^2(\pi_{u2p})^4(\pi_{g2p}^*)^4(\sigma_{u2p_z}^*)^2$	0		

Other Species Related to N₂

	O ₂	O ₂ ⁺	NO	NO ⁺	N ₂	CO	CN ⁻	C ₂ ²⁻
Electronic configuration	$\sigma^2\pi^4\pi^{*2}$	$\sigma^2\pi^4\pi^{*1}$	$\pi^4\sigma^2\pi^{*1}$	$\pi^4\sigma^2$	$\pi^4\sigma^2$	$\pi^4\sigma^2$	$\pi^4\sigma^2$	$\pi^4\sigma^2$
Bond order	2	2.5	2.5	3	3	3	3	3
Bond length (Å)	1.21	1.12	1.15	1.06	1.10	1.13	1.15	1.20
Bond energy (kJ/mol)	494	625	607	?	942	1070	891	839

Isoelectronic

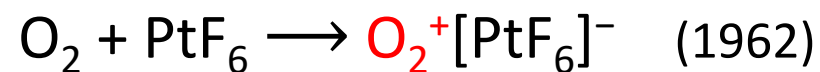
Isoelectronic

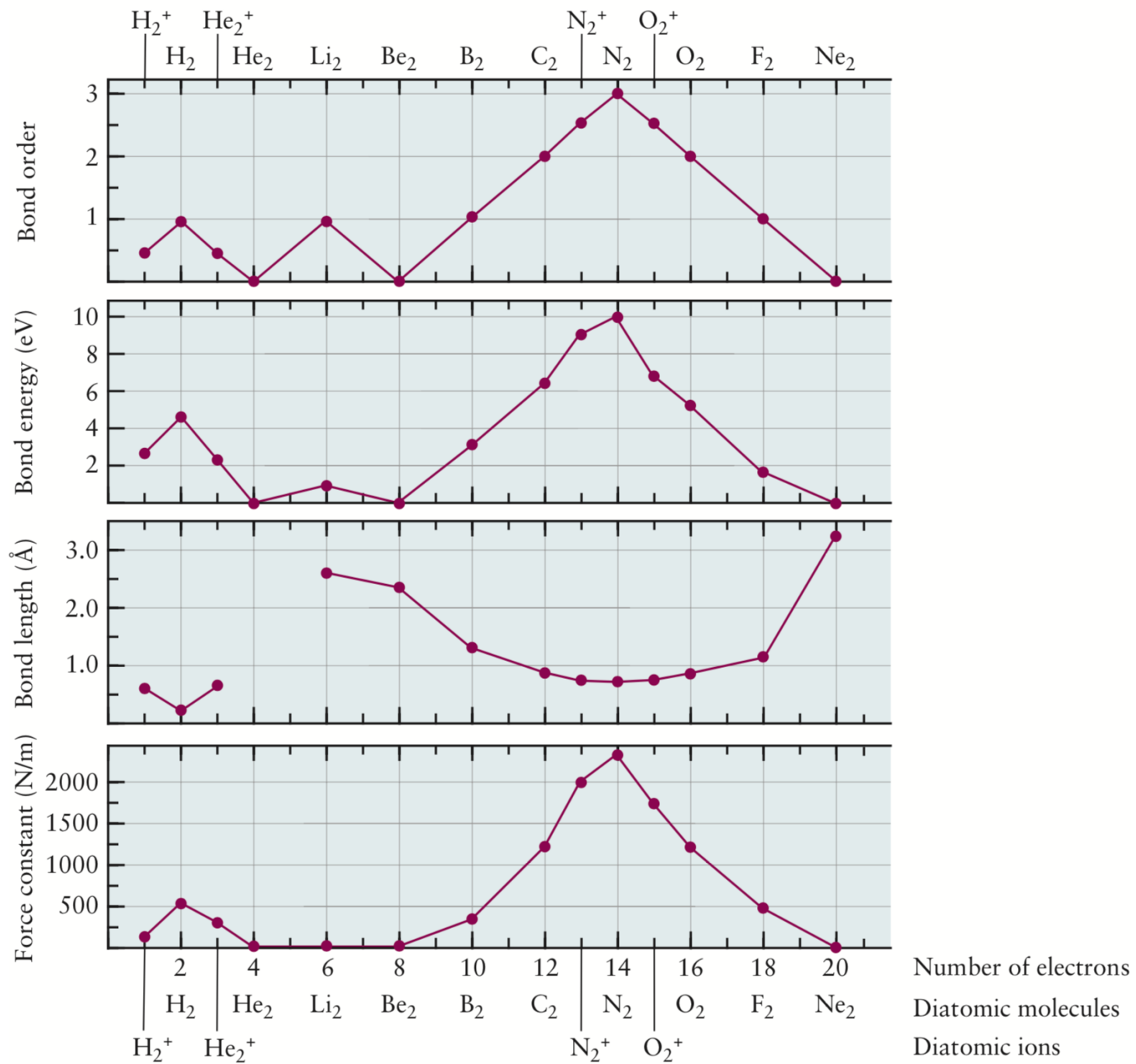


Other Species Related to O₂

	F ₂	O ₂ ²⁻	O ₂ ⁻	O ₂	O ₂ ⁺
Electronic configuration	$\sigma^2\pi^4\pi^{*4}$	$\sigma^2\pi^4\pi^{*4}$	$\sigma^2\pi^4\pi^{*3}$	$\sigma^2\pi^4\pi^{*2}$	$\sigma^2\pi^4\pi^{*1}$
Bond order	1	1	1.5	2	2.5
Bond length (Å)	1.41	1.49	1.33	1.21	1.12
Bond energy (kJ/mol)	154	142	~370	494	625

Isoelectronic





Determine the ground-state electron configuration and bond order of the F_2 molecule.

Major points of today's lecture

- **Quantum mechanics and molecular structure**

- 6. **Heteronuclear Diatomic Molecules**

- 7. **Summary Comments for the LCAO Method and Diatomic Molecules**

- 8. **Valence Bond Theory and the Electron Pair Bond**

- 9. **Orbital Hybridization for Polyatomic Molecules**

- 10. **Predicting Molecular Structures and Shapes**

- 11. **Using the LCAO and Valence Bond Methods Together**

- 12. **Summary and Comparison of the LCAO and Valence Bond Methods**

Heteronuclear Diatomic Molecules

1. A different set of labels: **heteronuclear diatomic molecules lack the inversion symmetry of homonuclear diatomic molecules.** (drop the g and u subscripts on the MO labels)

$$\sigma_{2s} = C_A 2s^A + C_B 2s^B$$

2. AOs on the participating atoms now correspond to **different energies: $C_A \neq C_B$**

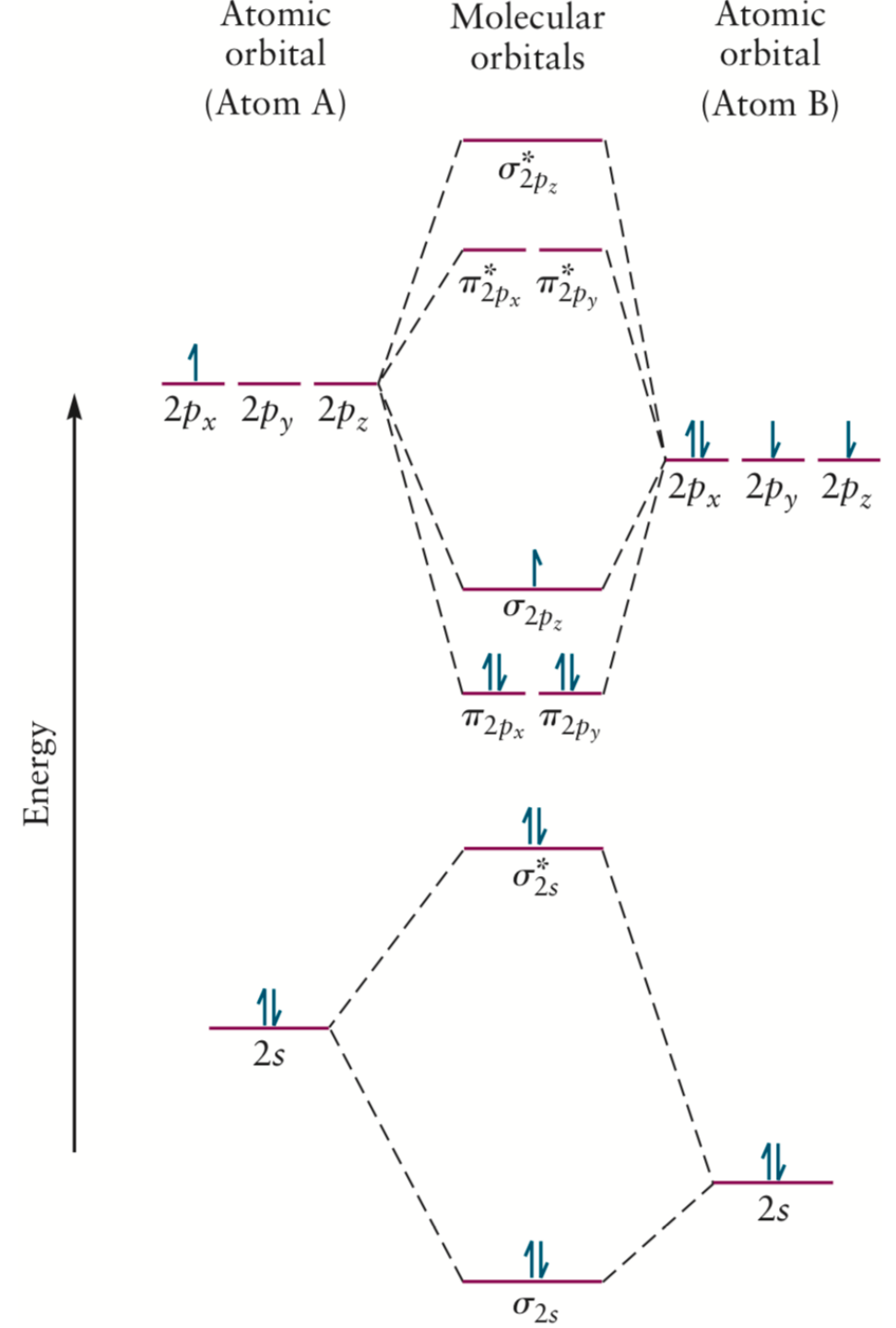
$$\sigma_{2s}^* = C'_A 2s^A - C'_B 2s^B$$

- Atom B is more electronegative than atom A: $C_B > C_A$ for the bonding σ MO (and the electron spends more time on the electronegative atom)
- $C'_A > C'_B$ for the higher energy σ^* MO and it more closely resembles a $2s^A$ AO

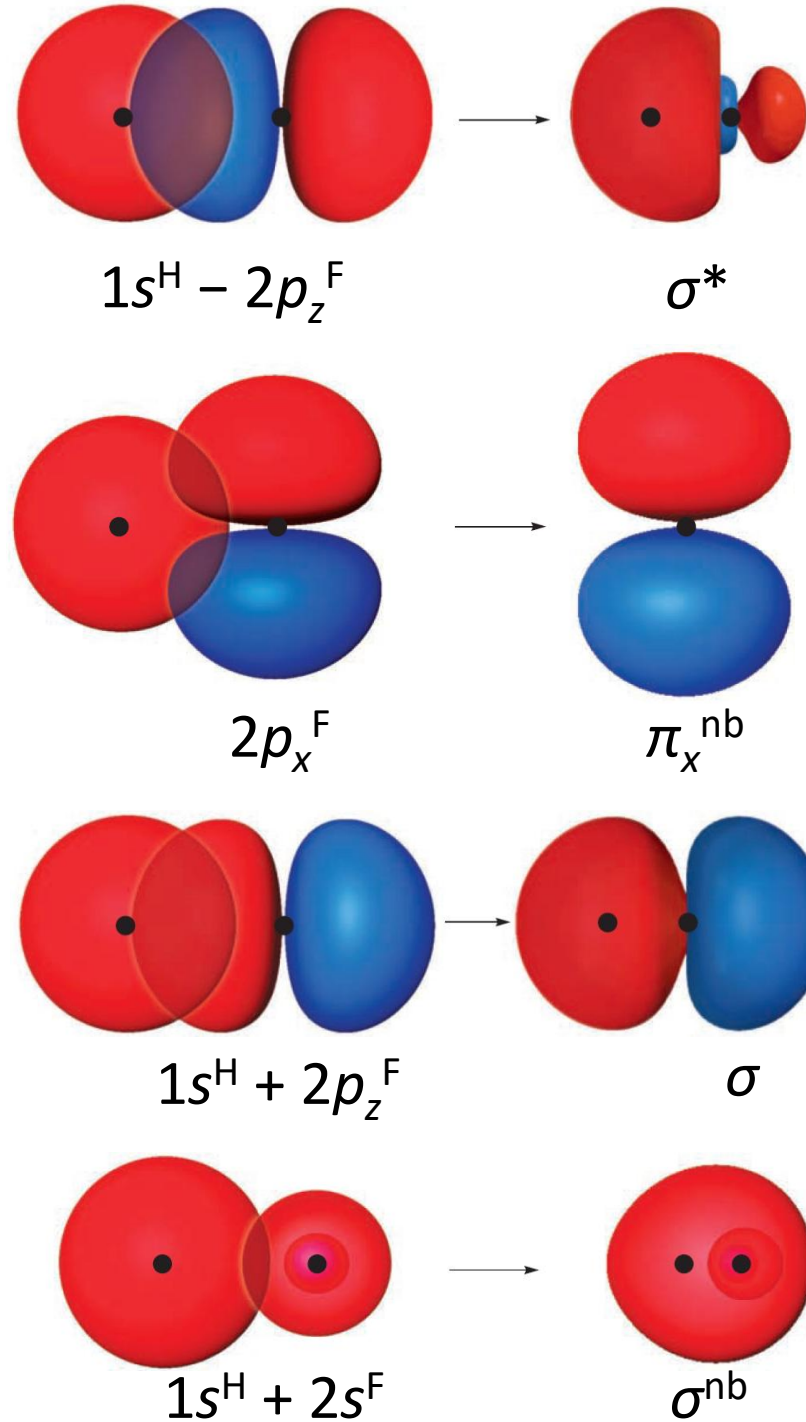
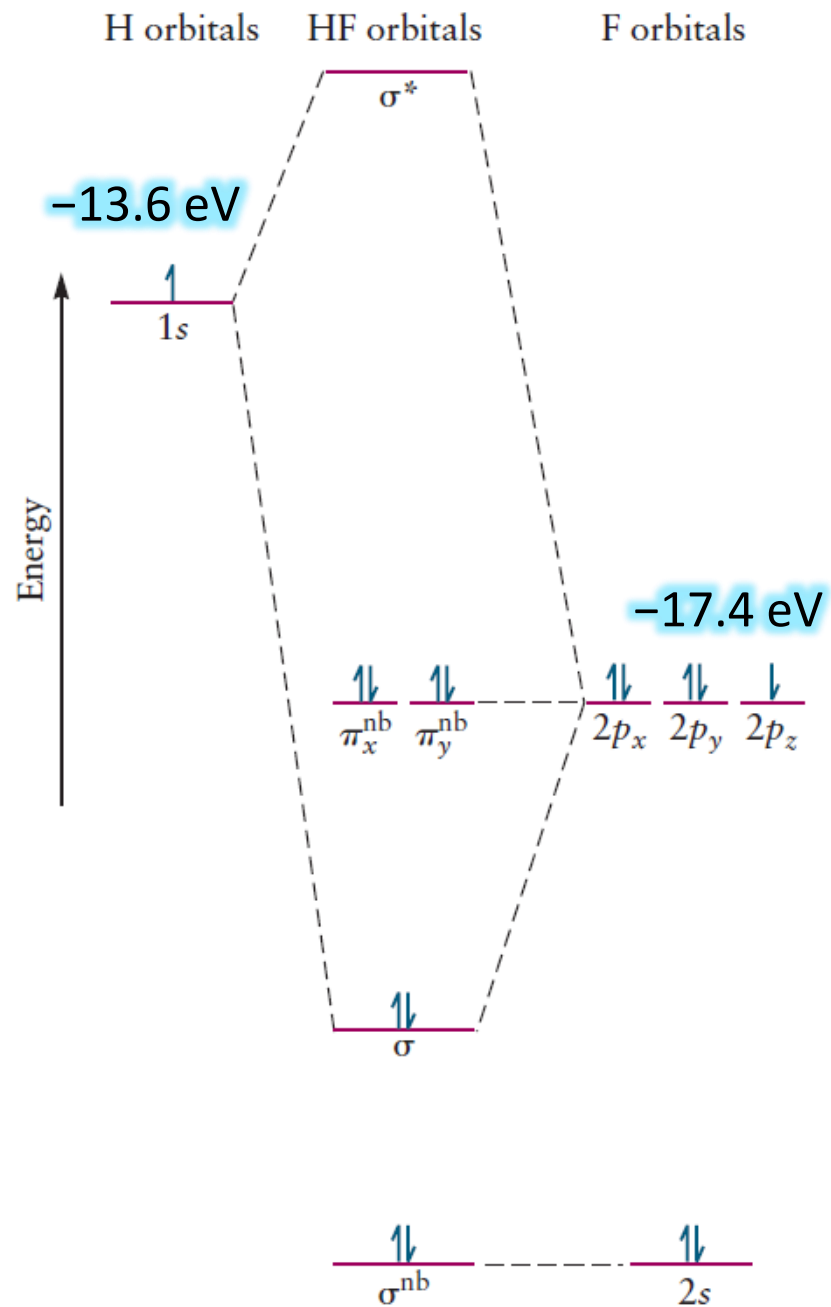
CO: carbon monoxide

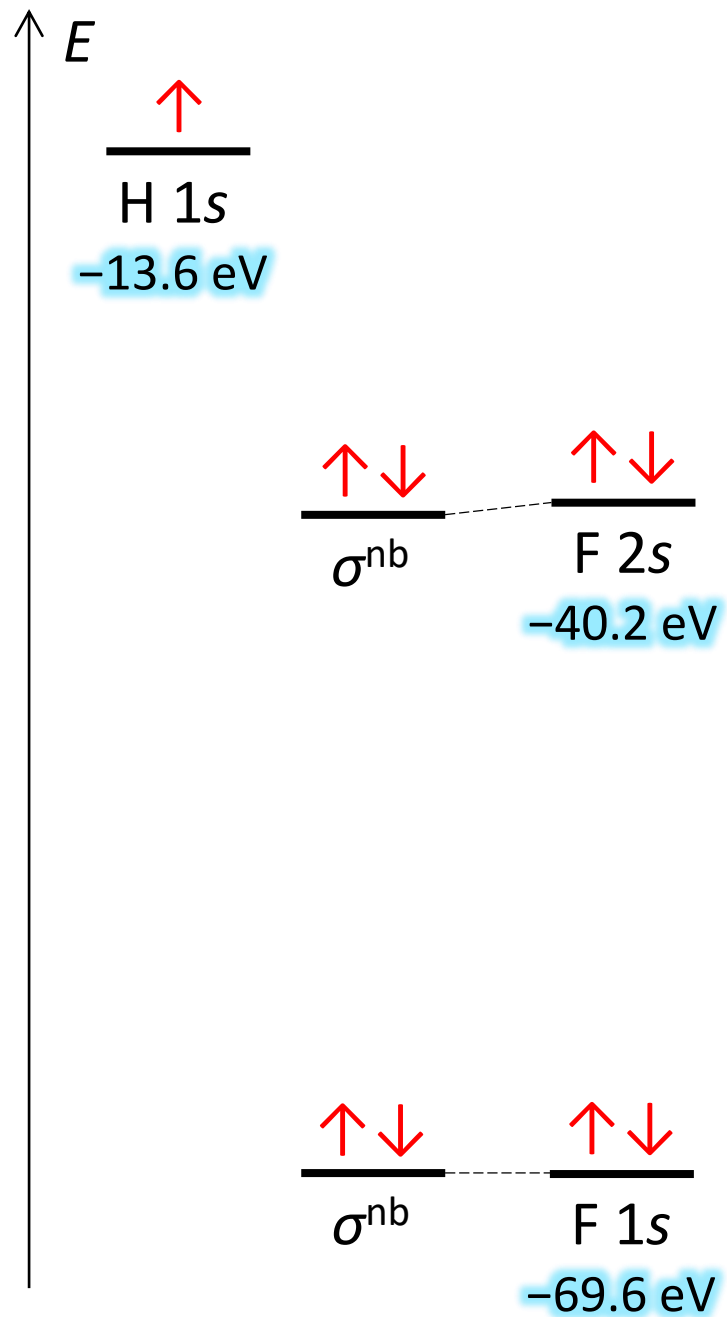
Heteronuclear Diatomic Molecules

FIGURE 6.20 Correlation diagram for heteronuclear diatomic molecules, AB. The atomic orbitals for the more electronegative atom (B) are displaced downward because they have lower energies than those for A. The orbital filling shown is that for (boron monoxide) BO.



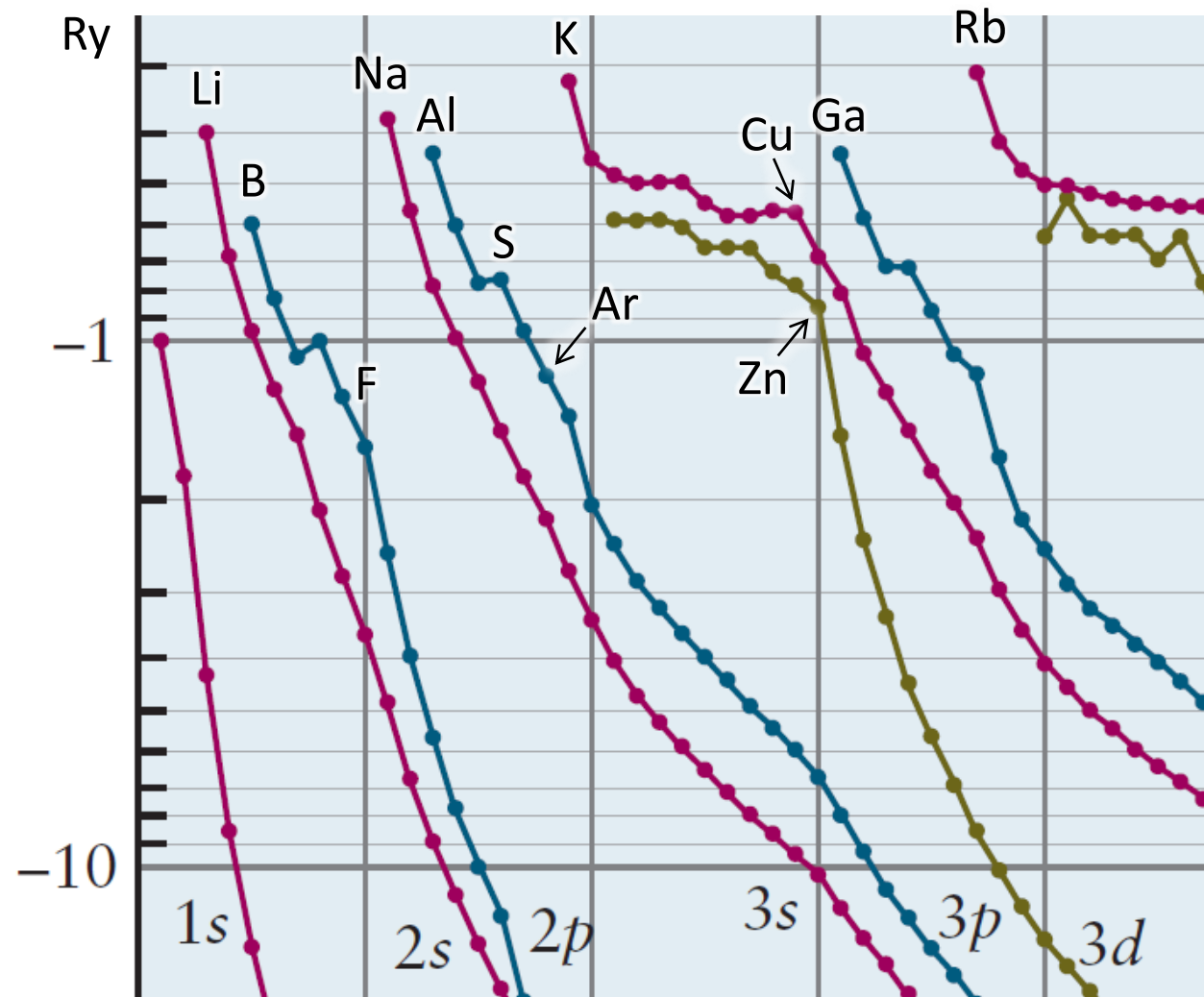
HF



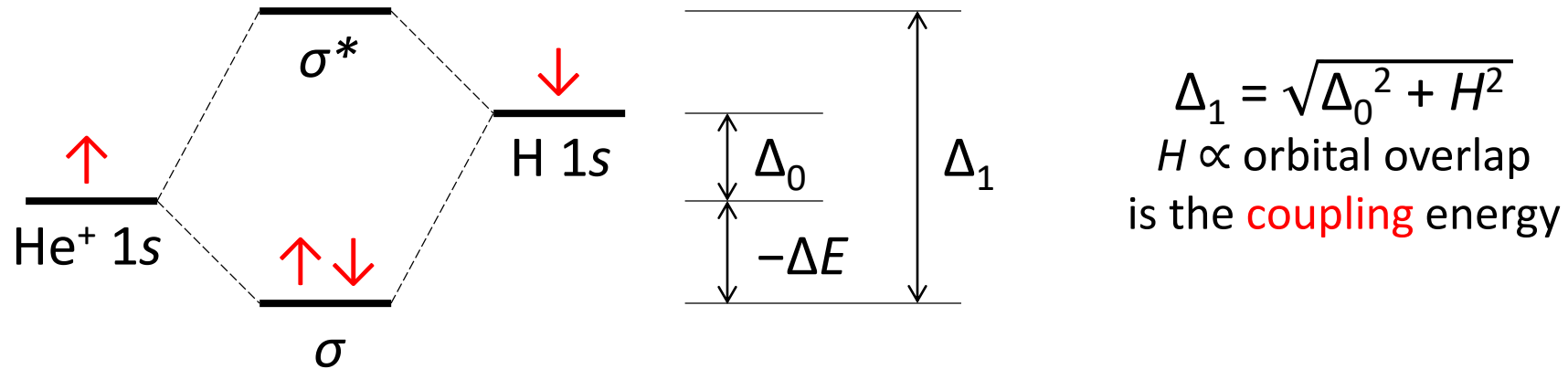


Unit:

Fig. 5.26, OGB8 p.210



A More General Correlation Diagram



$$-\Delta E = \frac{1}{2}(\Delta_1 - \Delta_0) = \frac{1}{2}(\sqrt{\Delta_0^2 + H^2} - \Delta_0)$$

Minimize

Maximize

Major points of today's lecture

- **Quantum mechanics and molecular structure**
 6. **Heteronuclear Diatomic Molecules**
 7. **Summary Comments for the LCAO Method and Diatomic Molecules**
 8. **Valence Bond Theory and the Electron Pair Bond**
 9. **Orbital Hybridization for Polyatomic Molecules**
 10. **Predicting Molecular Structures and Shapes**
 11. **Using the LCAO and Valence Bond Methods Together**
 12. **Summary and Comparison of the LCAO and Valence Bond Methods**

LCAO-MO: Summary

Atomic orbitals and molecular orbitals

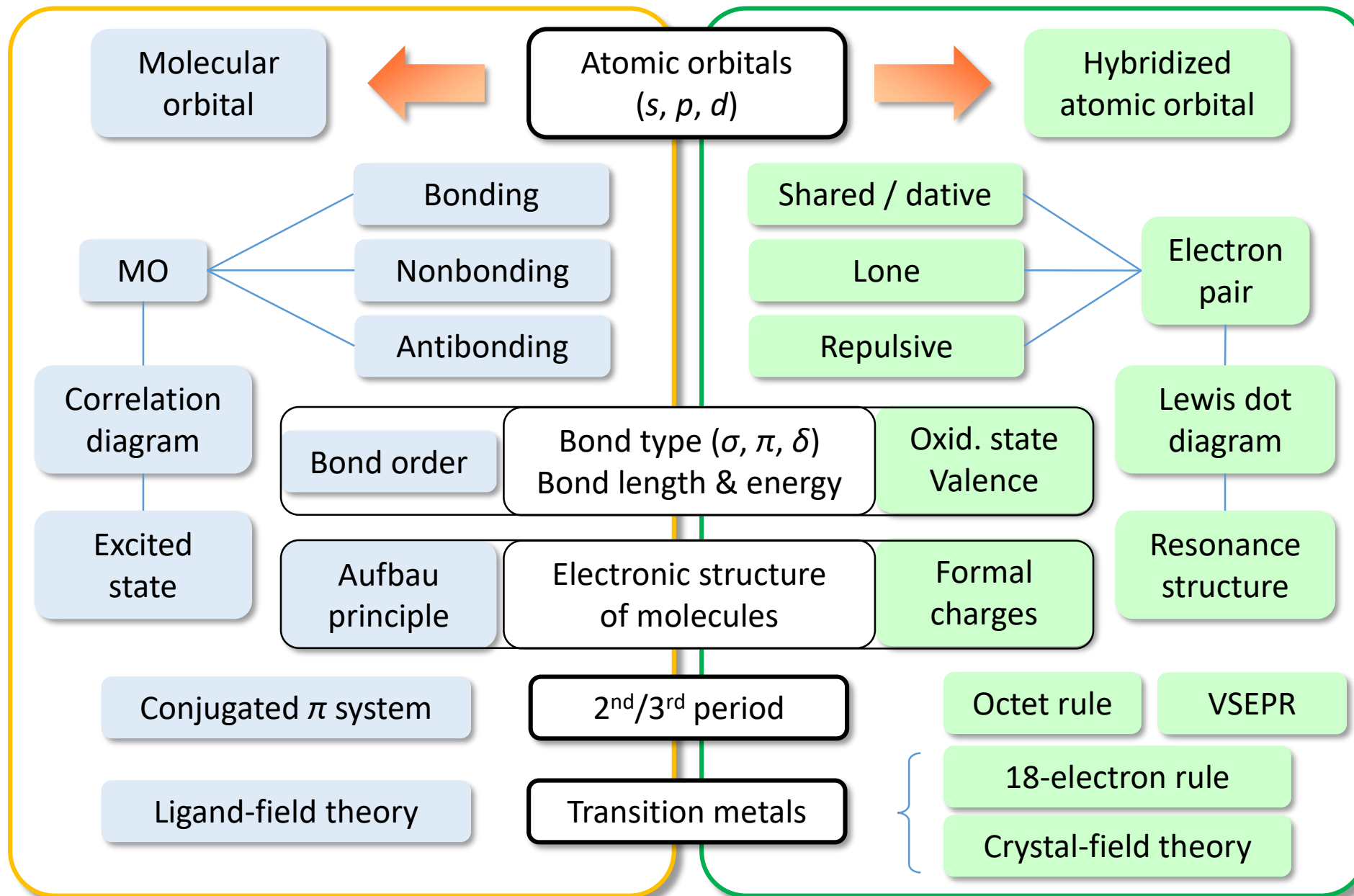
- Can be linearly combined
- Obey the Aufbau Principle

Rules of bonding

- Maximum orbital overlap
- Compatible orbital symmetries
- Approximate atomic orbital energies

Molecular Orbital (MO)

Valence Bond (VB) / Hybridization



Homework: read “principle of modern chemistry”, chapter 6.3; 6.4; 6.5